

Which one of the following is basic dimension?

- A. Velocity
- B. Energy
- C. Pressure
- D. Temperature

ANSWER: D

A _____ is defined as a quantity of matter or a region in space chosen for study.

- A. System
- B. Boundary
- C. Surrounding
- D. Mass

ANSWER: A

Which one of the following is not intensive property of a system?

- A. Temperature
- B. Pressure
- C. Density
- D. Total mass

ANSWER: D

Any change that a system undergoes from one equilibrium state to another is called a _____.

- A. System
- B. State
- C. Property
- D. Process

ANSWER: D

During a steady-flow process, fluid properties within the control volume may change with position but not with time.

- A. True
- B. False

ANSWER: A

Atmospheric pressure is measured by a device called a _____.

- A. Thermometer
- B. Manometer
- C. Barometer
- D. U-tube

ANSWER: C

Which one of the following is microscopic forms of energy?

- A. Kinetic energy
- B. Potential energy
- C. Internal energy
- D. Mechanical energy

ANSWER: C

_____ is the form of energy that is transferred between two systems (or a system and its surroundings) by virtue of a temperature difference.

- A. Mass
- B. Work
- C. Heat
- D. Temperature

ANSWER: C

ANSWER: A

A process during which there is no heat transfer is called _____.

- A. Isobaric process
- B. Isothermal process
- C. Isochoric process
- D. Adiabatic process

ANSWER: D

Energy can cross the boundary of a closed system in the form of _____.

- A. Heat or Mass
- B. Work or Mass
- C. Heat or Work
- D. Heat or Work or Mass

ANSWER: C

A liquid that is about to vaporize is called a _____.

- A. Saturated liquid.
- B. Compressed liquid.
- C. Saturated vapor.
- D. Sub cooled liquid.

ANSWER: A

ANSWER: B

The amount of energy absorbed or released during a phase-change process is called _____.

- A. Latent heat
- B. Thermal energy
- C. Work energy
- D. Heat energy

ANSWER: A

At pressures below the triple point value the process of passing from the solid phase directly into the vapor phase is called _____.

- A. Phase change
- B. Triple point
- C. Sublimation
- D. Melting

ANSWER: C

_____ the energy required to raise the temperature of a unit mass of a substance by one degree.

- A. Specific volume
- B. Specific enthalpy
- C. Specific heat
- D. Specific work

ANSWER: C

From the following statements, which one is Clausius's second law of thermodynamics?

- A. Energy can neither be created nor destroyed. It can only change forms. In any process, the total energy of the universe remains the same
- B. It is impossible for a self-acting machine to transfer heat from a colder body to a hotter one without the aid of an external agency
- C. It is impossible to design an engine that extracts heat and fully utilizes it into work without producing any other effect.
- D. As the temperature approaches absolute zero, the entropy of a system approaches a constant minimum.

ANSWER: B

A change in the state of gas during which the pressure of the gas remains constant is called _____ change.

- A. Adiabatic
- B. Isothermal
- C. Isobaric
- D. Isochoric

ANSWER: C

Which one of the following is ideal cycle for spark ignition reciprocating engines?

- A. Diesel cycle
- B. Otto cycle
- C. Rankine cycle
- D. Brayton cycle

ANSWER: B

25. Which method increases the thermal efficiency of simple ideal Rankine cycle on which steam power plant operates?

- A. Lowering the Condenser Pressure.
- B. Superheating the Steam to High Temperatures.
- C. Increasing the Boiler Pressure.
- D. All

ANSWER: D

28. Which one is not refrigeration cycle?

- A. Reversed Brayton cycle
- B. Ideal vapor compression refrigeration cycle
- C. Reversed Carnot cycle
- D. Brayton cycle

ANSWER: D

_____ the temperature at which condensation begins when the air is cooled at constant pressure.

- A. Adiabatic saturation temperature
- B. Dew point temperature
- C. Wet bulb temperature
- D. Dry bulb temperature

ANSWER: B

Which one of the following quantities is used to describe the amount of moisture the air holds relative to the maximum amount of moisture the air can hold at the same temperature?

- A. Specific humidity
- B. Relative humidity
- C. Absolute humidity
- D. Humidity ratio

ANSWER: B

Which one of the following is basic dimension?

- A. Velocity
- B. Energy
- C. Pressure
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ANSWER: D

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- C. Heat or Work
- D. Heat or Work or Mass

ANSWER: C

A liquid that is about to vaporize is called a _____.

- A. Saturated liquid
- B. Compressed liquid
- C. Saturated vapor
- D. Sub cooled liquid

ANSWER: A

Water boils at 100°C.

- A. True
- B. False

ANSWER: B

The amount of energy absorbed or released during a phase-change process is called_____.

- A. Latent heat
- B. Thermal energy
- C. Work energy
- D. Heat energy

ANSWER: A

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- B. Superheating the Steam to High Temperatures.
- C. Increasing the Boiler Pressure.
- D. All

ANSWER: D

Which term is related with the transfer of heat from a lower temperature region to a higher temperature one?

- A. Heat engine
- B. Work engine
- C. Refrigeration
- D. Thermal engine

ANSWER: C

_____ the temperature at which condensation begins when the air is cooled at constant pressure.

- A. Adiabatic saturation temperature
- B. Dew point temperature
- C. Wet bulb temperature
- D. Dry bulb temperature

ANSWER: B

Which one of the following quantities is used to describe the amount of moisture the air holds relative to the maximum amount of moisture the air can hold at the same temperature?

- A. Specific humidity
- B. Relative humidity
- C. Absolute humidity
- D. Humidity ratio

ANSWER: B

The buoyant force for a floating body passes through the:

- A. center of gravity of the body
- B. centroid of volume of the body
- C. meta-center of body
- D. centroid of displaced volume

ANSWER: D

Which of the following is the definition of dynamic viscosity?

- A. The force required to move a unit area of fluid parallel to a second unit area of fluid a unit

distance apart in a unit time.

- B. The ratio of the shear stress to the shear rate.
- C. The ratio of the dynamic viscosity to the density of the fluid.
- D. The time it takes for a known amount of fluid to flow a given distance.

ANSWER: A

Which of the following is the definition of kinematic viscosity?

- A. The force required to move a unit area of fluid parallel to a second unit area of fluid a unit distance apart in a unit time.
- B. The ratio of the shear stress to the shear rate.
- C. The ratio of the dynamic viscosity to the density of the fluid.
- D. The time it takes for a known amount of fluid to flow a given distance.

ANSWER: C

Which of the following is the best definition of dynamic viscosity?

- A. The resistance of a fluid to flow
- B. The ease with which a fluid flow
- C. The force required to move a fluid
- D. The velocity of a fluid

ANSWER: A

Which of the following is the best definition of kinematic viscosity?

- A. The resistance of a fluid to flow
- B. The ease with which a fluid flow
- C. The force required to move a fluid
- D. The velocity of a fluid

ANSWER: B

Which of the following is the best definition of density?

- A. The mass of an object per unit volume
- B. The volume of an object per unit mass
- C. The force of an object per unit volume
- D. The weight of an object per unit volume

ANSWER: A

Which of the following is the best definition of vacuum pressure?

- A. The pressure of a gas in a vacuum
- B. The pressure of a gas below atmospheric pressure
- C. The pressure of a gas above atmospheric pressure
- D. None of the above

ANSWER: B

Which of the following is a true statement about streamlines?

- A. Streamlines are smooth curves that follow the path of fluid particles.

- B. Streamlines are tangent to the velocity vector at every point.
- C. Streamlines never cross each other
- D. All of the above.

ANSWER: D

Which of the following is a characteristic of steady flow?

- A. The fluid particles move in smooth, parallel layers.
- B. The fluid particles move in a random, chaotic manner.
- C. The flow is steady, with the velocity at any point remaining constant over time.
- D. The flow is uniform, with the velocity at any point being the same in all directions.

ANSWER: C

Which of the following is the stream function?

- A. A function that gives the velocity potential of a fluid
- B. A function that gives the velocity of a fluid.
- C. A function that gives the pressure of a fluid.
- D. A function that gives the temperature of a fluid.

ANSWER: B

Which of the following is a vortex flow?

- A. A flow in which the fluid particles move in a circular path.
- B. A flow in which the fluid particles move in a straight line.
- C. A flow in which the fluid particles move in a random manner.
- D. A flow in which the fluid particles move in a wavy manner.

ANSWER: A

What is the Bernoulli principle?

- A. The pressure of a fluid increases as its velocity decreases.
- B. The pressure of a fluid decreases as its velocity increases.
- C. The pressure of a fluid is constant regardless of its velocity.
- D. The pressure of a fluid is inversely proportional to its velocity.

ANSWER: B

When heat is transferred from a hot body to a cold body in a straight line without affecting the intervening medium is known as-----.

- A. Conduction
- B. Convection
- C. Radiation
- D. Conduction and convection

ANSWER: C

Which one of the following are expected to have the highest thermal conductivity.

- A. Steam
- B. Solid ice

- C. Melting ice
- D. Water

ANSWER: B

The thermal conductivity of glass-wool varies from sample to sample b/c of.

- A. Composition
- B. Density
- C. Porosity
- D. All of the above

ANSWER: D

A steam pipe is to be insulated by two insulating materials put over each other.

- A. Better insulation should be put over the pipe and better one over it
- B. Interior simulation should be put over the pipe and a better one over it
- C. Both may be put in any order
- D. Whether to put interior oil over the pipe or the better one would depend on the steam temperature

ANSWER: A

The overall coefficient of heat transfer is used in case of-----.

- A. Conduction
- B. Convection
- C. Radiation
- D. Conduction and convection

ANSWER: D

-----Is the ratio of the emissive power and absorptive power of all bodies is the same and equal to the emissive power of a perfectly black body.

- A. Kirchoff's law
- B. Stefan's law
- C. Wen' law
- D. Planck's law

ANSWER: A

Which of the following property of air does not increase with the rise in temperature?

- A. Thermal conductivity
- B. Thermal diffusivity
- C. Dynamic viscosity
- D. Density

ANSWER: D

Which one of the following significantly affected by free convection heat transfer.

- A. Reynolds number
- B. Grashoff's number

- C. Prandtl number
- D. Stanton number

ANSWER: B

When a substance is heated and the temperature rises as the heat is added, the increase in heat is called _____

- A. sensible heat
- B. latent heat
- C. specific heat
- D. none of the above

ANSWER: A

What factors are to be considered for designing a heat exchanger-----

- A. Thermal analysis
- B. Hydraulic calculations
- C. Pressure calculations
- D. Thermal analysis and hydraulic calculations

ANSWER: D

Heat exchanger and condenser are same.

- A. True
- B. False

ANSWER: B

In shell and tube heat exchangers, the flow pattern is usually_____

- A. Concurrent
- B. Countercurrent
- C. Concurrent at a part and counter current at the other
- D. Mixed flow

ANSWER: C

The thermal conductivity of fin is should be _____with time.

- A. Exponentially increasing
- B. Increasing
- C. Decreasing
- D. Constant

ANSWER: D

Which one of the following heat exchanger is most efficient?

- A. Parallel flow
- B. Cross flow
- C. Counter flow
- D. Tangent flow

ANSWER: C

The flow of fluid caused by a pump, a fan, or by the atmospheric winds is called

- A. Natural flow
- B. Forced convection
- C. Free convection
- D. A and C

ANSWER: B

When heat is transferred from one particle to another by the actual motion of the heated particles is-----

- A. Conduction
- B. Convection
- C. Radiation
- D. Conduction and convection

ANSWER: A

The working cycle in case of four stroke engine is completed in following number of _____ revolutions of crankshaft.

- A. 1/2
- B. 1
- C. 2
- D. 4

ANSWER: C

Combustion in compression ignition engines is _____.

- A. Homogeneous
- B. Heterogeneous
- C. Both A and B
- D. Laminar

ANSWER: B

The presence of nitrogen in the products of combustion ensures that: _____.

- A. Complete combustion of fuel takes place
- B. Incomplete combustion of fuel occurs
- C. Dry products of combustion are analyzed
- D. Air is used for the combustion

ANSWER: D

The process of breaking up or a liquid into fine droplets by spraying is called as _____.

- A. Ionization
- B. Injection
- C. Atomization
- D. Carburetion

ANSWER: C

Black smoke appears in the exhaust due to _____.

- A. High load operation
- B. Rich Air – Fuel mixture
- C. Incomplete combustion
- D. combustion of lubrication oil

ANSWER: A

The ratio of indicated work done to the energy supplied by the fuel is called as _____.

- A. Thermal Efficiency
- B. Mechanical efficiency
- C. Volumetric efficiency
- D. Overall efficiency

ANSWER: A

A type of plant layout useful when the production process is in a continuous or repetitive way.

- A. Product or line layout
- B. Process or functional layout
- C. Fixed position or location layout
- D. Cellular layout

ANSWER: A

Is concerned with the achievement of optimal relationship between workers and their work environment.

- A. Organization
- B. management
- C. staff
- D. Ergonomics

ANSWER: D

One of the following is the function of forecasting.

- A. Forecasting sales
- B. Forecasting economic trend
- C. Forecasting staff needs
- D. All

ANSWER: D

A type of plant layout that is useful when the production process is organized in batches.

- A. Product or line layout
- B. Fixed position or location layout
- C. Process or functional layout
- D. Cellular layout

ANSWER: C

One of the following is not the characteristics of management.

- A. Management is universal
- B. Management is ongoing processes
- C. Management is goal oriented
- D. Management is individual activity

ANSWER: D

Which one of the following is true about forecasting.

- A. Forecast accuracy increases as time horizon decreases
- B. Forecasts are more accurate for groups
- C. Forecasts are not perfect
- D. All

ANSWER: D

Which one of the following are not the principle of good plant layout.

- A. Principle of integration
- B. Principle of minimum distance
- C. Principle of safety and satisfaction
- D. None

ANSWER: D

Is concerned with reducing, controlling and eliminating hazards from the industrial units.

- A. Industrial safety
- B. plant layout
- C. Ergonomics
- D. Inventory

ANSWER: A

Is the process of measuring, evaluating and monitoring the current activities how an organization has achieved its goals and to take any corrective actions needed to improve performance.

- A. Organizing
- B. Directing
- C. Staffing
- D. Controlling

ANSWER: D

Which type of organizational structure are temporary organizational structures formed for specific projects for a specific period of time and are dismantled, once the required goal is achieved.

- A. Matrix
- B. Functional
- C. Line & Staff

D. Line

ANSWER: A

Which one of the following factors are not internal affect productivity improvement.

A. Unhealthy working environment

B. Insufficient plant layout

C. Infrastructure supports

D. Poor maintenance system

ANSWER: C

Which one of the following statements is not true about productivity.

A. Higher productivity means to produce more with the same expenditure

B. Increase in production does necessarily by itself indicate increase in productivity.

C. Higher productivity means to produce more with a minimum increase in expense

D. Productivity is the efficient utilization of resources in producing goods/services.

ANSWER: B

Is the process of motivating, guiding, and supervising subordinates towards company objectives.

A. Organizing

B. Directing

C. Staffing

D. Controlling

ANSWER: B

Which one of the are not true about the importance of forecasting.

A. For long range planning

B. for marketing

C. For finance and accounting

D. None

ANSWER: D

Which range of forecasting time horizon, that used to forecast facility expansion and relocation.

A. Long range forecast

B. medium range forecast

C. Short range forecast

D. A & B

ANSWER: A

A type of layout that combines the resources necessary to manufacture a good or deliver a service, in one physical location.

A. Fixed position/project layout

B. Process layout

C. Combined layout

D. Product layout

ANSWER: A

Which type of organizational structure is difficult control because, there are no clear lines of authority and responsibility.

A. Matrix

B. Functional

C. Line & Staff

D. Line

ANSWER: B

Is the arrangement of industrial facilities such as personnel, operating equipment, storage space, materials handling equipment and all other supporting services.

A. Plant layout

B. Organization

C. Plant location

D. Ergonomics

ANSWER: A

----- is the time that elapses between issuing replenishment order and receiving the material in stores.

A. Replenishment time

B. Lead time

C. Idle-time

D. None of the above

ANSWER: B

Which of the following is false for inventory control?

A. Economic order quantity has a minimum total cost per order

B. Inventory carrying costs increases with an increasing quantity per order

C. Ordering cost decreases with an increasing quantity per order

D. Inventory carrying costs decrease with an increasing quantity per order

ANSWER: D

If an activity has multiple immediate predecessors, its earliest start (ES) is -----.

A. The maximum of all the earliest finishing values of its predecessors

B. Minimum latest starting of all immediately following activities

C. Latest finishing (LF) – activity time

D. Earliest starting (ES) + activity time

ANSWER: A

----- is the task of buying goods of the right quality in the right quantities, at the right time, and with the right price.

A. Supplying

- B. Purchasing
- C. Scrutinizing
- D. None of the above

ANSWER: B

----- is a type of operation in the network that neither requires any time nor any resources.

- A. Predecessor activity
- B. Dummy activities
- C. Successor activity
- D. Critical activities

ANSWER: B

Slack time is -----.

- A. $LF-ES/ES-LE$
- B. $ES-EF/EF-ES$
- C. $LF-EF/LS-ES$
- D. $LS-EF/EF-LS$

ANSWER: C

If an activity is an immediate predecessor for just a single activity, the Latest finishing (LF) equals the latest starting (LS) of the activity that immediately follows it.

- A. True
- B. False

ANSWER: A

Which of the following is not investment appraisal techniques.

- A. Payback period
- B. Accounting Rate of Return
- C. Economy ordering quantity model
- D. Profitability index

ANSWER: C

A machine costs of 1,000,000 birr, it produces items that generate a profit of 10 birr each on a production run of 50,000 units per year. What is the payback periods.

- A. 3 years
- B. 4 years
- C. 2 years
- D. 1 years

ANSWER: C

The primary goal of the plant layout is to minimize the profit by the arrangement of all the industrial facilities to the best advantage of total manufacturing of the product.

- A. True
- B. False

ANSWER: B

-----is the process of which planning, coordinates, supervises, controlling and carry out the tasks associated with the flow of materials; to, through, and out of an organization in an integrated manner.

- A. Materials Management
- B. Inventory Management
- C. Project Management
- D. Resource Management

ANSWER: A

Which of the following are the supplier sourcing.

- A. Single sourcing
- B. Sole sourcing
- C. Multiple sourcing
- D. All

ANSWER: D

-----is a stock of materials used to facilitate production or to satisfy customer demands. It includes raw-materials, work-in-progress and finished goods.

- A. Martials
- B. purchasing
- C. Inventory
- D. none

ANSWER: C

In which selective inventory control system, the base of classification is based on annual consumption value/items on the basis of their usage.

- A. ABC Analysis
- B. HML Analysis
- C. FSN Analysis
- D. VED Analysis

ANSWER: A

In which selective inventory control system, the base of classification is based on frequently pattern of items.

- A. ABC Analysis
- B. SDE Analysis
- C. FSN Analysis
- D. VED Analysis

ANSWER: C

In which selective inventory control system, the base of classification is based on lead time required to obtain the materials.

- A. HML Analysis
- B. SDE Analysis
- C. FSN Analysis
- D. VED Analysis

ANSWER: D

In which selective inventory control system, the base of classification is the nature of supplies.

- A. GOLF Analysis
- B. SDE Analysis
- C. SO-S Analysis
- D. VED Analysis

ANSWER: C

In which selective inventory control system, the base of classification is the source of supplies.

- A. GOLF Analysis
- B. SDE Analysis
- C. SO-S Analysis
- D. VED Analysis

ANSWER: A

Which one of the following is not true about Economic order quantity and inventory.

- A. Average inventory level and number of orders per year are inversely related.
- B. Ordering cost are non-linearly related to order size
- C. Carrying cost are linearly related to order size
- D. None

ANSWER: D

----- is a methodology for converting a large-scale project into detailed schedules for its thousands of activities for planning, scheduling and controlling purpose.

- A. Network Scheduling
- B. Work Breakdown Structure
- C. Bar Charts
- D. Milestone Charts

ANSWER: B

Which one of the following is not true about Critical Path Method Technique.

- A. Critical Path Method is mostly used in deterministic situations.
- B. Critical Path Method network is event-oriented
- C. Critical Path Method network is activity-oriented
- D. None

ANSWER: B

If an activity has only a single immediate predecessor, its earliest start (ES) equals the earliest finish (EF) of the predecessor.

- A. True
- B. False

ANSWER: A

-----is a method for shortening the project duration by reducing the time of one or more of the critical project activities to a time that is less than the normal activity time.

- A. Crash time
- B. Project crashing
- C. Crash cost
- D. Project risk

ANSWER: B

Which one of the following statements are not true.

- A. When the project time is reduced, the indirect cost decreases.
- B. When the project time is reduced, the direct cost increases.
- C. When the project time is reduced, the direct cost decreases.
- D. None

ANSWER: C

Which one of the following statements are not true

- A. The cost associated with risk events tends to be highest near the beginning of a project
- B. The probability of occurrence of risk events is highest near the beginning of a project
- C. The probability of occurrence of risk events is lowest near the end of a project
- D. The cost associated with risk events tends to be lowest near the beginning of a project

ANSWER: A

The following quantities are typically within the scope of mechanical measurements except

- A. Temperature
- B. Pressure
- C. Current
- D. Flow

ANSWER: C

Which of the following is fundamental quantity

- A. Time
- B. Pressure
- C. Velocity
- D. All

ANSWER: A

Which statement is false about primary standard?

- A. Are not available for use outside the national laboratories.
- B. Have highest accuracy and very expensive to own.

- C. Are used by the workers and technicians who actually carry out the measurements
- D. All are false

ANSWER: C

Indirect measurement involving one translation is called _____.

- A. Primary measurement
- B. Secondary measurement
- C. Tertiary measurement
- D. None

ANSWER: B

One of the following is not an advantage of digital device

- A. Accuracy
- B. High speed
- C. Decrease of human error
- D. No need of calibration

ANSWER: D

The revision applied to the critical value so that the final result obtained improves the worth of the result is called _____.

- A. Range
- B. Correction
- C. Error
- D. Actual value

ANSWER: B

The ratio of output response to a specified change in the input

- A. Sensitivity
- B. Readability
- C. Accuracy
- D. Repeatability

ANSWER: A

A pressure indicator showed a reading as 42 bar on a scale range of 0 - 50 bar. If the true value was 41.4 bar, what is the value of static error?

- A. 0.4 bar
- B. 0.6 bar
- C. 1.62 bar
- D. 1.45

ANSWER: B

For the above question number 8 what will be the value of percentage of relative error?

- A. 0.6
- B. 1.62

- C. 0.4
- D. 1.45

ANSWER: D

Which one of the following are not systematic error?

- A. Random errors
- B. Environmental errors
- C. Observational errors
- D. Instrumental errors

ANSWER: A

_____ Sensors translate movement into a change in the mutual inductance between magnetically coupled parts.

- A. Resistive
- B. Capacitive
- C. Inductive
- D. Ultrasonic

ANSWER: C

Identify the transducer not used for the measurement of temperature.

- A. Thermocouple
- B. Bourdon gage
- C. Thermometer
- D. Pyrometer

ANSWER: B

RTD and Thermistors are _____ transducers.

- A. Active and analog
- B. Passive and analog
- C. Active and digital
- D. Passive and digital

ANSWER: B

Which of the following is the input quantity provided to photoelectric type transducer?

- A. Pressure
- B. Light
- C. Sound
- D. Air

ANSWER: B

Which of the following properties a capacitive type transducer measures?

- A. Displacement
- B. Pressure
- C. Temperature

D. All the above

ANSWER: D

Which of the following device can be used for force measurement?

- A. Beams
- B. Bellows
- C. Capsule
- D. Bourdon tube

ANSWER: A

Which of the following can be measured using a Ring-type load cell?

- A. Large weigh
- B. Small weight
- C. Both large and small weights
- D. None of the mentioned

ANSWER: B

Which frequency is attenuated in a Low-Pass filter?

- A. High frequency
- B. Low frequency
- C. Mid-Range frequency
- D. No frequency

ANSWER: A

A device which is used to increase or augment the weak signal.

- A. Filter
- B. Amplifier
- C. Attenuator
- D. Transducer

ANSWER: B

Which of the following is an analog transducer?

- A. Encoders
- B. Strain gauge
- C. Digital tachometers
- D. Limit switches

ANSWER: B

Which of the following elements of control systems is responsible for transforming the output of a microprocessor/microcontroller into a controlling action on a machine?

- A. Sensor
- B. Actuator
- C. System

D. Controller

ANSWER: B

One of the following is different from the others

A. Diodes

B. Relays

C. Brushed DC Motor

D. Transistors

ANSWER: C

_____ is a mechanical actuator that give change of speed in mechatronic system.

A. Belt

B. Motors

C. Gears

D. Valves

ANSWER: C

Directional control valve is considered as

A. Finite Position

B. Infinite Position

C. ON/OFF Control

D. A & C

ANSWER: D

Produce force only in one direction by hydraulic/pneumatic pressure acting on the piston

A. Single acting cylinder

B. Double acting cylinder

C. Directional control valve

D. All

ANSWER: A

Which one is not the function of transducer in mechatronics?

A. To produce force only in one direction by hydraulic/pneumatic pressure acting on the Piston.

B. To make mechanically complex and expensive feeding and sorting devices

C. To provide identification and indication of the presence of different components

D. To act as protective mechanism for a system

ANSWER: B

The extent to which the measured value might be wrong and normally expressed in percentage is called

A. Error

B. Reproducibility

C. Sensitivity

D. Accuracy

ANSWER: D

_____ sensors rely on variation of the resistance of a material when the measured variable is applied to it.

- A. Resistive
- B. Capacitive
- C. Inductive
- D. Ultrasonic

ANSWER: A

An electronic component which is used as switching device

- A. Capacitor
- B. Inductor
- C. Resistor
- D. Transistor

ANSWER: A

Which of the following is velocity sensor?

- A. Tach generator
- B. Potentiometer
- C. Strain gage
- D. Orifice plate

ANSWER: B

A device which is used to increase or augment the weak signal.

- A. Filter
- B. Amplifier
- C. Attenuator
- D. Transducer

ANSWER: D

The term PLC stands for:

- A. Personal logic computer
- B. Programmable local computer.
- C. Personal logic controller
- D. Programmable logic controller.

ANSWER: D

One of the following is not the function of sequencing

- A. Sequencing
- B. Timing
- C. Logic
- D. None

ANSWER: D

An input condition that is open when de-energized describes

- A. Output
- B. Normally closed
- C. Normally open
- D. Negated Output

ANSWER: B

The ability of transducer to give the same output when used to measure a constant input over a period of time is called _____

- A. Stability
- B. Sensitivity
- C. Reproducibility
- D. Repeatability

ANSWER: A

All are displacement sensor except

- A. Strain-gauged element
- B. Potentiometer
- C. Capacitive element
- D. Piezoelectric

ANSWER: D

A filter that has a pass band which allows all frequencies from zero up to some frequency to be transmitted.

- A. High pass filter
- B. Low pass filter
- C. Band stop filter
- D. Band pass filter

ANSWER: B

The Wheatstone bridge can be used to convert

- A. Resistance changes to a voltage change
- B. Current change to a resistance change
- C. Voltage changes to a resistance change
- D. Voltage changes to a current change

ANSWER: A

_____ is a motor that produces rotation through equal angles for each digital pulse supplied to its input.

- A. Servo motor
- B. Ac motor
- C. DC motor
- D. Stepper motor

ANSWER: A

A combination of components which are coordinated together to perform a specific objective.

- A. Modeling
- B. System
- C. Mechatronics
- D. All

ANSWER: B

Which one are not methods of reducing stress concentration?

- A. Avoiding sharp corners.
- B. Providing fillets.
- C. Use of multiple holes instead of single hole
- D. None

ANSWER: D

Which of the following is not the function of a spring?

- A. They are used for the measurement of force and to control motion
- B. They are used to absorb shocks and vibrations
- C. They are used to store energy
- D. They are used to measure displacement

ANSWER: D

Longitudinal stress in a cylinder is given by [symbols have their usual meanings].

- A. $PD/2t$
- B. $2PD/t$
- C. $PD/4t$
- D. $4PD/t$

ANSWER: C

If the spring is compressed completely and the adjacent coils touch each other, the length of spring is called as?

- A. Solid length
- B. Compressed length
- C. Free length
- D. None of the mentioned

ANSWER: A

Find the mean coil diameter of a helical compression spring if a load of 1200N is applied on the spring. Spring index is 6, and wire diameter 7mm.

- A. 7/6mm
- B. 42mm
- C. $1200 \times 6/7$ mm
- D. None of the listed

ANSWER: B

Which one of the following are a measure of stiffness?

- A. Modulus of elasticity
- B. Modulus of plasticity
- C. Resilience
- D. Toughness

ANSWER: A

Which of the following is not a type of welding joint?

- A. Lap joint
- B. Butt joint
- C. Tee joint
- D. Single strap joint

ANSWER: D

In which of the following case stress concentration factor is ignored?

- A. Ductile material under static load
- B. Ductile material under fluctuating load
- C. Brittle material under +static load
- D. Brittle material under fluctuating load

ANSWER: A

Use of multiple notches in a V shaped flat plate will

- A. Reduce the stress concentration
- B. Increase the stress concentration
- C. No effect
- D. Cannot be determined

ANSWER: A

An unfired pressure vessel is used to carry steam, gases or fluids at pressure.

- A. Less than atmospheric pressure
- B. Greater than atmospheric pressure
- C. Equal to atmospheric pressure
- D. None of the mentioned

ANSWER: B

Which of the following theory of failure is mostly used for ductile materials?

- A. Maximum principal stress theory
- B. Maximum shear stress theory
- C. Distortion energy theory
- D. Haigh's theory

ANSWER: B

The phenomenon of decreased resistance of the materials to fluctuating stresses is the main

characteristic of _____ failure.

- A. Fracture
- B. Fatigue
- C. Yielding
- D. None of the mentioned

ANSWER: B

The shaft is always stepped with _____ diameter at the middle portion and _____ diameter at the shaft ends.

- A. Minimum, maximum
- B. Maximum, minimum
- C. Minimum, minimum
- D. Zero, infinity

ANSWER: B

Which of the following act on shafts?

- A. Torsional moment
- B. Bending Moment
- C. Both torsional and bending
- D. None of the mentioned

ANSWER: C

Which of the following can be used for power transmission in intersecting shafts?

- A. Spur Gear
- B. Helical Gear
- C. Bevel Gear
- D. None of the listed

ANSWER: C

A _____ bearing supports the load acting along the axis of the shaft.

- A. Thrust
- B. Radial
- C. Longitudinal
- D. Transversal

ANSWER: A

Modulus of resilience is defined as

- A. Strain energy per unit volume
- B. Strain energy per unit area
- C. Independent of strain energy
- D. None of the mentioned

ANSWER: A

If driving gear rotates ACW, driven gear will rotate in which direction?

- A. ACW
- B. CW
- C. Both CW & ACW
- D. Can't be determined

ANSWER: A

Which of the following is not true about gears?

- A. Positive drive
- B. Constant velocity ratio
- C. Transmit large power
- D. Bulky construction

ANSWER: D

Which of the following is not a type of sunk key?

- A. Gib headed key
- B. Woodruff key
- C. Square key
- D. Hollow saddle key

ANSWER: D

Which one of the following is the first most important aspect of reliability?

- A. Design
- B. Production
- C. Transportation
- D. All

ANSWER: A

Several findings show the reason why seals fail. How many percent does improper installation constitutes to the failure?

- A. 12 percent
- B. 15 percent
- C. 26 percent
- D. 22 percent

ANSWER: C

All but one is not the way of achieving a more reliability.

- A. To have fewer number of components
- B. To have a series system
- C. To have a parallel system
- D. To have backup component

ANSWER: B

The cost of spare parts is more than ____ of the total maintenance cost in the industry.

- A. 45
- B. 50%

C. 55%

D. 60%

ANSWER: B

The following are the most commonly used inventory analyses except one.

A. FSM

B. SDE

C. VED

D. HML

ANSWER: A

In which generation, the maintenance was restricted to fixing the components when it breaks because it was the cheapest alternative.

A. Second Generation

B. Third Generation

C. First generation

D. Fourth generation

ANSWER: C

Which of the following are not maintenance objectives.

A. Reducing downtime.

B. Minimizing energy usage.

C. Improving equipment efficiency

D. Minimizing production

ANSWER: D

Which of the following are true about the purpose of planned maintenance

A. To prevent the occurrence of breakdown

B. If breakdown has occurred then to restore it to original condition

C. If breakdown has occurred then discard the items

D. A & b

ANSWER: D

----- is done to keep an equipment or machinery in a satisfactory operating condition through regular inspection, calibration, lubrication or replacement of certain components.

A. Preventive Maintenance

B. Corrective Maintenance

C. Predictive Maintenance

D. Running Maintenance

ANSWER: A

Which of the following are the category of Preventive Maintenance

A. Shut-down Maintenance

B. Design-out Maintenance

- C. Running Maintenance
- D. All

ANSWER: D

Which of the following are the causes of equipment breakdown?

- A. Unsound foundation
- B. Too high or too low voltage supply
- C. Lack of lubrication
- D. All

ANSWER: D

-----Is a system that utilizes to collected data from the machinery in order to predict and prevent the potential failures?

- A. Intelligent Maintenance System
- B. Normal Maintenance System
- C. Typical Maintenance System
- D. Old Maintenance System

ANSWER: A

Damage is influenced by:

- A. environmental conditions
- B. conditions of use of equipment
- C. operational processes
- D. All

ANSWER: D

Damage is classified as:

- A. Corrosion
- B. Wear
- C. Fatigue
- D. All

ANSWER: D

Failures are resulting from:

- A. within the operation
- B. material or information input
- C. customer actions
- D. All

ANSWER: D

The damage in sliding bearings is the result of:

- A. Wear plus Fatigue
- B. Wear
- C. Fatigue

D. None

ANSWER: A

Due to damages, bearings will either gets:

A. Heated up

B. Abnormal noise

C. Vibration

D. All

ANSWER: D

Which of the following are the defects rolling bearing?

A. Bearing inner race loose on shaft

B. Bearing running dry

C. Miss-alignment

D. All

ANSWER: D

How much of the annual steel production is lost due to corrosion.

A. 5%

B. 3%

C. 10%

D. 15%

ANSWER: A

Which of the following are the causes of shaft failure?

A. Wear

B. Fatigue

C. Misalignment

D. All

ANSWER: D

Which of the following remedies are true for preventing shaft failures?

A. Wear of can be prevented by using proper lubrication

B. Fatigue can be reduced by proper design of local areas

C. Misalignment can be eliminated by proper assembly procedures

D. All

ANSWER: D

Which of the following are the objectives of accurate alignment?

A. Minimize the amount of shaft bending

B. Minimize the amount of wear

C. Reduce excessive axial and radial forces

D. All

ANSWER: D

Which of the following are the symptoms of misalignment?

- A. High casing temperatures at or near the bearings
- B. Loose foundation bolts
- C. Excessive amount of oil leakage at the bearing seals
- D. All

ANSWER: D

The state of damage of equipment is depends on:

- A. Kind of damage
- B. condition of damage
- C. Duration of operation of equipment
- D. All

ANSWER: D

-----Is the determination of the technical state or state of damage of equipment?

- A. Maintenance technique
- B. Technical diagnostics
- C. Sampling techniques
- D. None

ANSWER: B

-----is the measuring technique of the state of damage that determines the difference between the initial state and the state after wear process.

- A. Direct measured
- B. Indirect measured
- C. Absolute measurement
- D. Related measurement

ANSWER: A

-----are techniques of condition monitoring that mostly carried out without interruption of operation of the unit.

- A. On-load monitoring techniques
- B. Off-load monitoring techniques
- C. Dynamic monitoring techniques
- D. Particle monitoring techniques

ANSWER: A

Most failures give some warning before they occur. This warning is-----

- A. functional failure
- B. Potential failure
- C. Physical failure
- D. Chemical failure

ANSWER: B

-----is defined as an identifiable physical condition which indicates that a functional failure is either about to occur or it is in the process of occurring.

- A. Potential failure
- B. functional failure
- C. Physical failure
- D. Chemical failure

ANSWER: A

-----is the interval between the occurrence of a potential failure and its deterioration in to functional failure.

- A. P-F interval
- B. P-D interval
- C. P-S interval
- D. None

ANSWER: A

Which of the following are true about P-F interval?

- A. warning period
- B. lead time to failure
- C. The failure development period.
- D. All

ANSWER: D

Which of the following are categories of condition monitoring techniques?

- A. Dynamic monitoring
- B. Temperature monitoring
- C. Chemical monitoring
- D. All

ANSWER: D

Which of the following are the general-purpose monitoring techniques?

- A. Lubricant monitoring
- B. Thermal monitoring
- C. Vibration/noise monitoring
- D. All

ANSWER: D

Which of the following are the basic elements of maintenance technology?

- A. Operation
- B. Servicing
- C. Repair.
- D. All

ANSWER: D

-----are a type of decisions that have been encountered and made in the past.

- A. Programmed decisions
- B. Non programmed decisions
- C. Associative choices
- D. None

ANSWER: A

----- are a type of decision-making environments, it has more than one states of nature exist but lacks sufficient knowledge and insufficient information to know the consequences of different actions.

- A. Decision Making under Conditions of Certainty
- B. Decision Making under Conditions of Risk
- C. Decision Making under Conditions of Uncertainty
- D. None

ANSWER: C

----- are a type of decision-making environments, it has more than one states of nature exist but it has sufficient information to know the consequences of different actions.

- A. Decision Making under Conditions of Certainty
- B. Decision Making under Conditions of Risk
- C. Decision Making under Conditions of Uncertainty
- D. None

ANSWER: B

-----Is the probability that an item/equipment or system will perform its intended function for a stated period of time under specified operating conditions?

- A. Maintainability
- B. Availability
- C. Reliability
- D. None

ANSWER: C

As a product has more components, what will be the reliability?

- A. The reliability is decrease
- B. The reliability is increase.
- C. The reliability is constant
- D. All

ANSWER: A

Which of the following are not true about debugging phase in hazard rate curves?

- A. Hazard rate is rapid decrease
- B. The Weibull distribution $\beta < 1$
- C. Hazard rate is constant

D. Infant mortality

ANSWER: C

Which of the following are not true about wear-out phase in hazard rate curves?

A. End of life failure

B. hazard rate is increases

C. Weibull distribution $\beta > 1$

D. Weibull distributions $\gamma = 1$

ANSWER: D

Which of the following are not true about useful life phase in hazard rate curves?

A. hazard rate is constant

B. Weibull distributions $\gamma = 1$

C. the failures occur randomly manner

D. Weibull distribution $\beta < 1$

ANSWER: D

Which of the following are true about the relationship between Weibull analysis and hazard rate?

A. When $B = 1$, the hazard function is constant

B. When $B < 1$, we get a decreasing hazard function and

C. When $B > 1$, we get increasing hazard function

D. All

ANSWER: D

Which one of the following are the commonly measure of availability.

A. Inherent Availability

B. Achieved Availability

C. Operational Availability

D. All

ANSWER: D

----- is the probability that items or system can be retained in, or can be restored to operating condition in a specified amount of time.

A. Maintainability

B. Availability

C. Reliability

D. None

ANSWER: A

For the successful spare parts management, it is essential to analyze the spare parts inventory based on:

A. Frequency of issues

B. Annual consumption value

- C. Criticality
- D. All

ANSWER: D

Which of the following are the objectives of spare parts managements?

- A. Maximizing spare parts availability,
- B. Minimize environmental costs
- C. Minimizing the economic costs.
- D. All

ANSWER: D

Maintenance schedule can be prepared at:

- A. Long-range schedule
- B. Weekly schedule
- C. Daily schedule
- D. All

ANSWER: D

Which of the following are maintenance schedule techniques?

- A. Modified Gantt chart
- B. CPM
- C. PERT
- D. All

ANSWER: D

Which of the following are the reconditioning steps?

- A. Inspection
- B. Repair
- C. Overhaul
- D. All

ANSWER: D

----- Is the process by which jobs are matched with resources and sequenced to be executed at a certain point in time?

- A. Scheduling
- B. Planning
- C. Networking
- D. Staffing

ANSWER: A

Which of the following are true about maintenance planning?

- A. Reduced maintenance cost
- B. Improved utilization of the maintenance workforce
- C. Improved quality of maintenance work

D. All

ANSWER: D

Which of the following are true about maintenance scheduling?

- A. Facilitates proper sequence in maintenance service
- B. Eliminates chances of sudden breakdown
- C. Facilitates optimum use of highly paid maintenance staff
- D. All

ANSWER: D

Which of the following are true about objectives of planning and scheduling maintenance?

- A. Minimizing the idle time of maintenance workers
- B. Maximizing the efficient use of work time, material, and equipment
- C. Maintaining the operating equipment at a responsive level in terms of delivery schedule and quality.
- D. All

ANSWER: D

What would happen, if equipment possesses reliability and maintainability to the maximum extent in accordance to MTTR?

- A. Failure rate is higher and downtime is longer
- B. Failure rate is lower and downtime is longer
- C. Failure rate is higher and downtime is shorter
- D. Failure rate is lower and downtime is shorter

ANSWER: D

The _____ goes on increasing with the increase in degree of maintenance efforts.

- A. Cost of down time
- B. Cost of spares and maintenance
- C. Labor and overhead cost
- D. All of the above

ANSWER: C

With the increase in preventive maintenance cost, breakdown maintenance cost

- A. Increases
- B. Decreases
- C. Remains same
- D. Any of the above

ANSWER: B

A systematic approach for maintenance is?

- A. Problem – Cause – Diagnosis –Rectification
- B. Problem – Diagnosis - Cause – Rectification
- C. Problem – Measure – Diagnosis –Rectification

D. Problem – Diagnosis – Measure –Rectification

ANSWER: A

MTTR is estimated to:

- A. Determine the actual repair work required
- B. Determine work maintenance
- C. Calculate labor cost
- D. Estimate the time required in marketing

ANSWER: A

All but one is not the way of achieving a good reliability.

- A. To have fewer number of components
- B. To have backup components
- C. To have a series system
- D. To have a parallel system

ANSWER: C

Which of the following is defined as average of the time taken in repair work?

- A. Preventive maintenance
- B. Total productive maintenance
- C. Predictive maintenance
- D. Breakdown maintenance

ANSWER: A

The cost of spare parts is more than _____ of the total maintenance cost in the industry.

- A. 45
- B. 50%
- C. 55%
- D. 60%

ANSWER: B

This type of schedule allows 10% to 15% of the workforce to be available for emergency work.

- A. Long-range
- B. Medium-range
- C. Weekly
- D. Daily

ANSWER: C

Which one of the following is not true about Proper machine installations?

- A. Minimizing maintenance challenge
- B. Maximize life cycle costs of production machinery
- C. Maximize capacity utilization
- D. Maximize reliability

ANSWER: B

In designing the foundation for an alignment-critical machine, all factors are important except:

- A. Different foundation block
- B. Stiffness of a foundation
- C. Local soil conditions
- D. Foundation thickness

ANSWER: A

One of the following reinforcement and anchoring approaches provides relatively a more rigid support.

- A. Leveling Wedges
- B. Leveling Screws
- C. Anchor Bolts and Shims
- D. All

ANSWER: A

Which one of the following is the first most important aspect of reliability?

- A. Design
- B. Production
- C. Transportation
- D. All

ANSWER: A

Which one of the following is not the benefit of predictive type of maintenance

- A. Low maintenance cost
- B. Less machine failure
- C. Low production
- D. Longer machine life

ANSWER: C

Is a set of activities that detect changes in the physical condition of equipment (signs of failure) in order to carry out the appropriate maintenance work.

- A. Preventive maintenance
- B. Predictive maintenance
- C. Reactive maintenance
- D. Breakdown maintenance

ANSWER: B

Which of the following is an example of preventive maintenance?

- A. Replacing a broken machine part
- B. Cleaning an air filter
- C. Repairing a malfunctioning machine
- D. None of the above

ANSWER: B

Which of the following is a benefit of preventive maintenance?

- A. Increased downtime
- B. Increased repair costs
- C. Increased equipment life
- D. None of the above

ANSWER: C

Which of the following is not a type of preventive maintenance?

- A. Inspections
- B. Lubrication
- C. Calibration
- D. Replacing broken parts

ANSWER: D

The P-F curve shows all but not one

- A. Where failure in the equipment starts to occur.
- B. Where equipment condition deteriorates to the point at which the failure can be detected; and finally.
- C. Where the equipment has failed indicating functional failure.
- D. None

ANSWER: D

On-condition tasks are carried out at

- A. less than the P-F intervals
- B. greater than the P-F intervals
- C. at the P-F intervals
- D. A and C

ANSWER: A

Which of the following is a technique used to monitor the condition of rotating machinery?

- A. Oil analysis
- B. Acoustic emission testing
- C. Vibration analysis
- D. Thermography

ANSWER: C

Which of the following is a technique used to monitor the condition of bearings?

- A. Oil analysis
- B. Acoustic emission testing
- C. Thermography
- D. Partial discharge testing

ANSWER: A

Which of the following is a technique used to monitor the condition of insulation in electrical

equipment?

- A. Oil analysis
- B. Acoustic emission testing
- C. Thermography
- D. Partial discharge testing

ANSWER: C

Which of the following is a common type of vibration sensor?

- A. Microphone
- B. Thermocouple
- C. Accelerometer
- D. Pressure transducer

ANSWER: C

Which of the following is NOT a factor considered when determining spares provisioning levels?

- A. Lead time
- B. Consumption value
- C. Production capacity
- D. Criticality of the equipment

ANSWER: C

Which of the following is NOT a goal of spares provisioning?

- A. Minimizing downtime
- B. Maximizing repair costs
- C. Reducing inventory costs
- D. Ensuring availability of necessary spare parts

ANSWER: B

Which of the following is NOT a factor to consider when determining the appropriate level of spares provisioning?

- A. The cost of the spare parts
- B. The frequency of equipment failures
- C. The lead time required to obtain spare parts
- D. The age of the equipment

ANSWER: D

What is the primary goal of improving availability?

- A. To reduce the number of repairs needed
- B. To increase the utilization of the equipment or system
- C. To reduce maintenance costs
- D. To increase the number of spare parts available

ANSWER: B

Which of the following is a common metric used to measure maintainability?

- A. Mean Time Between Failure (MTBF)
- B. Mean Time To Repair (MTTR)
- C. Overall Equipment Effectiveness (OEE)
- D. None of the above

ANSWER: B

What is the primary goal of improving maintainability?

- A. To reduce the number of repairs needed
- B. To increase the utilization of the equipment or system
- C. To reduce maintenance costs
- D. To reduce the time required to perform repairs

ANSWER: D

Which of the following is NOT a characteristic of a maintainable equipment or system?

- A. Easy access to components that require maintenance
- B. Standardized components and procedures
- C. High complexity
- D. Diagnostic capabilities

ANSWER: C

Which of the following is NOT a step in the maintenance planning process?

- A. Identifying maintenance needs
- B. Conducting maintenance activities
- C. Developing a maintenance schedule
- D. Assigning maintenance personnel

ANSWER: C

Which of the following is a common tool used in maintenance engineering?

- A. Failure Modes and Effects Analysis (FMEA)
- B. Root Cause Analysis (RCA)
- C. Pareto Analysis
- D. All of the above

ANSWER: D

What is maintenance technology?

- A. The process of performing maintenance on equipment or systems
- B. The tools and techniques used to support maintenance activities
- C. The process of measuring the effectiveness of maintenance activities
- D. The process of scheduling maintenance activities

ANSWER: B

Which one of the following is NOT the advantage of standardization?

- A. Improves in reliability
- B. Eliminates the need for special components

- C. Supports “cannibalizing” maintenance approaches
- D. Increases maintenance time and cost

ANSWER: D

What is the life history curve of equipment?

- A. A graph that shows the cost of equipment over time
- B. A graph that shows the performance of equipment over time
- C. A graph that shows the maintenance requirements of equipment over time
- D. A graph that shows the failure rate of equipment over time

ANSWER: C

Ultra-Sonic Machining removes materials using the _____ tool.

- A. Perpendicularly rotating
- B. Inclined oscillating
- C. Perpendicularly oscillating
- D. Axially oscillating

ANSWER: D

Additive manufacturing uses much less material than other subtractive manufacturing processes.

- A. true
- B. false

ANSWER: A

Which of the following is typically the cheapest type of 3D printer?

- A. FDM
- B. Powder based
- C. SLA
- D. SLM

ANSWER: A

What is the function of flux in submerged arc welding?

- A. To completely cover the welded zone
- B. To prevent oxidation of joint
- C. To prevent spattering of molten metal
- D. To prevent sticking of molten metal

ANSWER: D

In which of the following gas welding process a non-consumable electrode is used?

- A. Submerged arc welding
- B. Tungsten inert gas welding
- C. Stud welding
- D. Gas metal arc welding

ANSWER: B

In plasma arc welding the gas is?

- A. Ionized
- B. Heated
- C. Magnetized
- D. Vaporized

ANSWER: A

Which of the following gas welding process uses constant voltage?

- A. Submerged arc welding
- B. Tungsten inert gas welding
- C. Stud welding
- D. Gas metal arc welding

ANSWER: D

In a single point turning operation with a cemented carbide and steel combination having a Taylor exponent of 0.25, if the cutting speed is halved, then the tool life will become

- A. Eight times
- B. Sixteen times
- C. Half
- D. Two times

ANSWER: B

SLA printing uses a plastic strand that's pushed through a heated nozzle.

- A. True
- B. False

ANSWER: B

In the Taylor's tool life equation, $VT^n = C$, the value of $n=0.5$. The tool life has a life of 180 minutes at a cutting speed of 18 m/min. If the tool life is reduced to 45 minutes, then the cutting speed will be

- A. 36 m/min
- B. 72 m/min
- C. 18 m/min
- D. 9 m/min

ANSWER: A

3D printing technology is expanding and is now able to print metal parts.

- A. True
- B. False

ANSWER: B

Which of the following is typically the most expensive type of 3D printer?

- A. SLA
- B. FDM
- C. SLM
- D. None

ANSWER: C

From the following printers which one can print multiple materials at one time?

- A. FDM
- B. SLA
- C. SLM
- D. None

ANSWER: A

Gas welding is also known as

- A. oxyacetylene welding
- B. oxy welding
- C. oxy-fuel welding
- D. all of the above

ANSWER: D

An acetylene/oxygen flame burn at ____ °C.

- A. 2250
- B. 2526
- C. 3073
- D. 3773

ANSWER: D

From the following which printer melts metal?

- A. SLS
- B. SLA
- C. SLM
- D. FDM

ANSWER: C

Which type of 3D printer uses a pool of resin to create the solid part?

- A. FDM
- B. SNL
- C. SLA
- D. None

ANSWER: C

Which of the following process includes heating of green compact at high temperature in the powder metallurgy process?

- A. Sintering

- B. Compacting
- C. Scaling
- D. Atomization

ANSWER: A

The welder uses the ____ flame as the starting point for all other flame adjustments.

- A. Reducing
- B. Neutral
- C. Oxidizing
- D. None

ANSWER: B

Which one among the following welding processes uses non-consumable electrode?

- A. Gas metal arc welding
- B. Submerged arc welding
- C. Gas tungsten arc welding
- D. Flux coated arc welding

ANSWER: C

In oxidizing flame, the color of inner cone is

- A. light blue
- B. dark blue
- C. purplish tinge
- D. red

ANSWER: C

Which of the following is (are) property(ies) of ceramic?

- A. Brittle
- B. Heat resistant
- C. Corrosion resistant
- D. All

ANSWER: D

Which of the following is ceramic forming technique(s)?

- A. injection moulding
- B. slip casting
- C. shaping by hand
- D. all of the above

ANSWER: D

The process in which molten metal is forced to pass through a nozzle in to a steam of water or air is called_____

- A. Electrolysis
- B. Shotting

C. Atomization

D. Reduction

ANSWER: C

Thermoplastics is an example of

A. Carbon and Graphitic matrix composites

B. Ceramic matrix composites

C. Metal matrix composites

D. Polymer matrix composites

ANSWER: D

Composites are used in

A. long span bridges

B. power generating wind mills

C. earthquake proof highway supports

D. all of the above

ANSWER: D

which one is the function of a matrix?

A. Holds the fibers together

B. Protects the fibers from environment

C. Enhances transverse properties of a laminate

D. All of the above

ANSWER: D

Which one of the following secondary operations is used to improve the self-lubricating capacity of the sintered part in powder metallurgy?

A. Infiltration

B. Impregnation

C. Repressing

D. Coining

ANSWER: B

Which one of the following methods is not used for producing metal powders?

A. Atomization

B. Machining and grinding

C. Compacting

D. Electrolysis

ANSWER: C

Following is the earliest known fibers used to reinforce materials

A. Glass fibers

B. Carbon fibers

C. Plant fibers

D. Wood fibers

ANSWER: A

From the following which one is the sequence of operations in conventional powder metallurgy process?

- A. Atomization__Blending and mixing__Compacting__Sintering
- B. Atomization__Blending and mixing__Sintering__Compacting
- C. Blending and mixing__Compacting__Atomization__Sintering
- D. Blending and mixing__Atomization__Compacting__Sintering

ANSWER: A

What material is not used in 3D printing?

- A. Nylon
- B. PLA
- C. ABS
- D. PVC

ANSWER: D

Which file type is most commonly exported from CAD software?

- A. SLDRT
- B. STL
- C. JPG
- D. All

ANSWER: B

What does SLS stands for?

- A. Selective laser sintering
- B. Separated light sintering
- C. Selective lithographic solution
- D. None of the above

ANSWER: A

The first step in each product design process is to _____ and conceptualize the function and appearance of the product.

- A. Summarize
- B. Visualize
- C. Imagine
- D. None

ANSWER: B

In manufacturing self-lubricating bearings by powder metallurgy an important secondary operation that is carried out after sintering is

- A. Cold isostatic pressing
- B. Impregnation

- C. Hot isostatic pressing
- D. Infiltration

ANSWER: B

Spherical metal powders are produced by:

- A. Electrolytic process
- B. Atomization
- C. Oxidation
- D. Reduction

ANSWER: B

Which technology requires a filling pattern to be produced within the enclosed boundary?

- A. Binder jetting
- B. Vat polymerization
- C. Material extrusion
- D. None of the above

ANSWER: C

Fused deposition modeling is a layer AM process that uses a _____ filament by fused deposition.

- A. Thermoplastic
- B. Polylactic
- C. Thermoelastic
- D. None

ANSWER: A

Which of the following method is used to make powder for brittle metals?

- A. Mechanical pulverization
- B. Chemical reduction
- C. Electrolytic process
- D. Atomization

ANSWER: A

Which method is used to make powder of metals having low melting point?

- A. Mechanical pulverization
- B. Chemical reduction
- C. Electrolytic process
- D. Atomization

ANSWER: D

LOM machines are primarily used for rapid prototyping of _____

- A. Metallic parts
- B. Wooden parts
- C. Ceramic parts

D. Plastic parts

ANSWER: D

Which of the following components influence the material removal rate in Abrasive jet machining?

A. Nozzle

B. Abrasive grain

C. Carrier gas

D. All of the mentioned

ANSWER: D

Sintering increases

A. Electrical conductivity, density and ductility

B. Electrical conductivity, density and brittleness

C. Porosity, electrical conductivity and brittleness

D. Porosity, density and ductility

ANSWER: A

What is the major problem in hot extrusion?

A. Design of punch

B. Design of die

C. Wear and tear of die

D. Wear of punch

ANSWER: B

This is a bulk forming process in which the workpiece or billet is shaped into finished part by the application of compressive and tensile forces with the help of a pair of rods.

A. Rolling

B. Extrusion

C. Forging

D. Wire drawing

ANSWER: C

In heading, the maximum length that can be upset in a single blow is _____ the diameter of the initial wire stock.

A. Two times

B. Four times

C. One times

D. None

ANSWER: D

The heat generated in resistance welding is expressed by

A. $2I^2RT$

B. I^2RT

- C. IRT2
- D. 4I2RT

ANSWER: B

What type of electrodes are used in resistance seam welding

- A. Flat
- B. Disk
- C. Domed
- D. Pointed

ANSWER: D

Which of the following process would produce strongest components?

- A. Hot rolling
- B. Extrusion
- C. Cold rolling
- D. Forging

ANSWER: C

Casting defects due to metal starts freezing before reaching the center of the casting for a casting with two gates at its two sides is called _____

- A. Pin hole
- B. Cold shut
- C. Mis run
- D. Gas holes

ANSWER: B

In the following properties of nozzle, which of them does not influence the MRR in AJM?

- A. Size
- B. Outside temperature
- C. Wear
- D. Distance from work piece

ANSWER: B

What is the cause of pinhole defects in casting?

- A. Metal starts freezing before reaching the farthest point in the cavity
- B. Entrapped spherical shaped bubbles
- C. Tiny blow holes occurring either at or just below the casting surface.
- D. Metal starts freezing before reaching the center of the casting for a casting with two gates at its two sides

ANSWER: C

Density in sintering product is improved by

- A. Hot pressing
- B. Cold pressing

- C. Coining
 - D. A and C
- ANSWER: D

Friction at the tool-chip interface can be reduced by

- A. Decreasing the rake angle
- B. Decreasing the cutting speed
- C. Increasing the depth of cut
- D. Increasing the cutting speed

ANSWER: D

The main drawback to the use of FDM technology is_____.

- A. Quality
- B. Speed of construction
- C. Surface finish
- D. All

ANSWER: B

In traditional machining, which component is made of a relatively harder material?

- A. Tool
- B. Workpiece
- C. Holding device
- D. None of the mentioned

ANSWER: A

For machining to take place, tool should be penetrated into work piece to a certain _____

- A. Depth
- B. Height
- C. Length
- D. Width

ANSWER: A

Which motion is responsible to determine the geometry of jobs to be machined?

- A. Only Motion of tool
- B. Only motion of work piece
- C. Relative motion between tool and work piece
- D. None of the mentioned

ANSWER: C

For large positive back rake angle, tool will be

- A. Weaker
- B. Stronger
- C. Smoother
- D. Harder

ANSWER: A

Which of the following will give better chip flow?

- A. Positive back rake angle tool
- B. Negative back rake angle tool
- C. Zero back rake angle tool
- D. None of the mentioned

ANSWER: A

Which of the following tools are generally manufactured by Powder metallurgy?

- A. Cemented carbides
- B. High carbon steel
- C. Abrasives
- D. Low carbon steel

ANSWER: A

Angle between side cutting edge and axis of tool is known as

- A. Side rake angle
- B. Side relief angle
- C. Side cutting edge angle
- D. Back rake angle

ANSWER: C

What must be the velocity of the carrier gas that carries the abrasive particles in AJM?

- A. 10 – 50 m/sec
- B. 300-500 m/sec
- C. 50 – 150 m/sec
- D. 150 –300 m/sec

ANSWER: D

Which one of the following processes does NOT use laser?

- A. LOM
- B. SLS
- C. SLA
- D. FDM

ANSWER: D

The tool life increases with the

- A. Decrease in back rake angle
- B. Increase in side cutting edge angle
- C. Decrease in side rake angle
- D. Decrease in nose radius

ANSWER: B

Which of the following cutting conditions greatly affects the tool wear?

- A. Depth of cut
- B. Feed
- C. Cutting speed
- D. None of the above is correct.

ANSWER: C

In abrasive jet machining, as the distance between the nozzle tip and the work surface increases, the material removal rate

- A. Increases continuously
- B. Decreases continuously
- C. Decreases becomes stable and then increases
- D. Increases, becomes stable and then decreases

ANSWER: D

The defects due to metal starts freezing before reaching the farthest point in the cavity is:

- A. Cold shut
- B. Mis run
- C. Pin hole
- D. Gas holes

ANSWER: B

Which of the following materials can be machined using ECM?

- A. Hard nonconductive materials
- B. Hard conductive materials
- C. All conductive materials
- D. All of the above

ANSWER: B

After debarring is done, which of the following is formed?

- A. Flat surfaces
- B. Controlled radius
- C. Sharp edges
- D. All

ANSWER: B

Which casting defect occurs due to excess moisture content in molding sand?

- A. Porosity
- B. Blow holes
- C. Shrinkage
- D. Inclusion

ANSWER: B

Gas hole defect is the result of:

- A. Entrapped spherical shaped bubbles
 - B. Metal starts freezing before reaching the farthest point in the cavity
 - C. Tiny blow holes occurring either at or just below the casting surface
 - D. Metal starts freezing before reaching the center of the casting for a casting with two gates at its two sides
- ANSWER: A

Which type of sand is used in shell molding?

- A. Black sand
- B. Any of the sand
- C. Wet and fine sand
- D. Dry and fine sand

ANSWER: D

In which one of the following ionized gas are used to cut metal sheet?

- A. Ultrasonic machining
- B. Laser beam machining
- C. Plasma arc machining
- D. Electron beam machining

ANSWER: C

Pattern used in shell moldings is normally made up of?

- A. Wax
- B. Plastic
- C. Metal
- D. Wood

ANSWER: C

In order to generate a casting with an internal surface, which part is typically used in a casting operation?

- A. Chaplets
- B. Core
- C. Riser
- D. Pattern

ANSWER: B

The property by virtue of which a sand mold is capable of with standing a high temperature of the molten metal with out fusing is known as_____

- A. Porosity
- B. Cohesiveness
- C. Adhesiveness
- D. Refractoriness

ANSWER: D

Hollow casting is the other name of which of the following special casting process?

- A. Slush casting
- B. Precision investment casting
- C. Vacuum die casting
- D. Squeeze casting

ANSWER: A

Which of the type of centrifugal casting methods is used to produce 'non-symmetrical' shaped castings?

- A. True centrifugal casting
- B. Centrifuge casting
- C. Semi centrifugal casting
- D. None centrifugal casting

ANSWER: D

In which one of the following sand casting the bonding of the mold process takes place without heating?

- A. Cold box
- B. No bake
- C. Dry sand mold
- D. A and B

ANSWER: D

From the following which one is the advantage of shell molding?

- A. Capability to make thin cross sections in casting
- B. Mold collapsibility usually avoids cracks in casting
- C. Pattern need not be removed from the mold
- D. The casting requires minimal finishing and cleaning operations

ANSWER: B

Advanced machining process by which small portion of the worksurface to be removed by fusion or vaporization is:

- A. Mechanical machining
- B. Electro chemical machining
- C. Thermal machining
- D. Chemical machining

ANSWER: C

Types of machining used to remove round sharp corners on holes in metal parts produced by conventional through-hole drilling is called:

- A. Electro chemical drilling
- B. Electro chemical debarring
- C. Electro chemical grinding
- D. None of the above

ANSWER: B

One of the following casting processes uses polystyrene pattern:

- A. Investment casting
- B. Lost foam
- C. Lost wax process
- D. All

ANSWER: D

In which one of the following casting process oil bonded sand and gray iron are used as a typical core material?

- A. Squeeze casting
- B. Semisolid metal casting
- C. Permanent mold casting
- D. Slush casting

ANSWER: C

From the following which one falls under the trade name full mold process?

- A. Squeeze casting
- B. Semi-solid metal casting
- C. Slush casting
- D. Evaporative pattern casting

ANSWER: D

Tool life is measured by the

- A. Number of pieces machined between tool sharpening's
- B. Time the tool is in contact with the job
- C. Volume of material removed between tool sharpening's
- D. All of the above

ANSWER: D

In machining metals, surface roughness is due to

- A. Feed marks or ridges left by the cutting tool
- B. Fragment of built-up edge on the machined surface
- C. Cutting tool vibrations
- D. All of these

ANSWER: D

Which of the following statement is correct for orthogonal cutting system?

- A. The cutting edge of the tool is perpendicular to the direction of tool travel
- B. The cutting edge clears the width of the workpiece on either ends
- C. The chip flows over the tool face and the direction of the chip flow velocity is normal to the cutting edge
- D. All of the above

ANSWER: D

Crater wear leads to

- A. Increase in cutting temperature
- B. Weakening of tool
- C. Friction and cutting forces
- D. All of these

ANSWER: D

Drilling is an example of

- A. Orthogonal cutting
- B. Oblique cutting
- C. Simple cutting
- D. Uniform cutting

ANSWER: B

Cast iron during machining produces

- A. Continuous chips
- B. Discontinuous chips
- C. Continuous chips with built-up edge
- D. B & C are correct

ANSWER: B

In metal machining, the zone where the heat is generated due to friction between the moving chip and the tool face, is called

- A. Friction zone
- B. Work-tool contact zone
- C. Shear zone
- D. None of these

ANSWER: A

Thermal cracking of tools occurs at

- A. Low temperature
- B. High temperature
- C. Low cutting speed
- D. None of the mentioned

ANSWER: B

In the orthogonal cutting of metals _____

- A. the cutting edge of the tool is perpendicular to the direction of tool travel
- B. the cutting forces occur in one direction only
- C. the cutting edge is wider than the depth of cut
- D. all of the mentioned

ANSWER: A

By increasing cutting speed, amount of heat generated

- A. Increases
- B. Decreases
- C. Remains constant
- D. None of the mentioned

ANSWER: A

From the following, which one is not independent variable in cutting process?

- A. tool material and coatings
- B. temperature rise in the workpiece, the tool, and the chip,
- C. cutting fluids
- D. cutting speed, feed, and depth of cut

ANSWER: B

In forging, specifically warm working, the working temperature is _____.

- A. $0.3 T_m$
- B. $0.1 T_m$
- C. $0.5 T_m$
- D. $0.75 T_m$

ANSWER: A

A property of sand due to which it evolves a great amount of steam and other gasses is called

- A. permeability
- B. cohesiveness
- C. adhesiveness
- D. none

ANSWER: A

Green sand is a mixture of

- A. 30% sand & 70% clay
- B. 70% sand 30% clay
- C. 50% sand & 50% clay
- D. 90% sand & 10% clay

ANSWER: B

The cold working of metals is carried out by

- A. at a recrystallization temperature
- B. the any temperature
- C. below the recrystallization temperature
- D. above the recrystallization temperature

ANSWER: C

The process of cutting hole of various shapes and sizes in to a work piece

- A. trimming
- B. punching
- C. slitting
- D. die

ANSWER: B

Cold working of a metals increase

- A. tensile strength
- B. hardness
- C. yield strength
- D. all of these

ANSWER: D

Which one of the following engineering materials is defined as a compound containing metallic and nonmetallic elements:

- A. ceramic
- B. metal
- C. composite
- D. polymer

ANSWER: C

Which of the following processes start with a material that is in a fluid or semifluid state and solidifies the material in a cavity:

- A. casting
- B. machining
- C. forging
- D. pressing

ANSWER: A

Which one of the following is a machine used to perform extrusion:

- A. forge hammer
- B. rolling mill
- C. milling machine
- D. press

ANSWER: D

Which of the following materials is corrosive?

- A. zinc oxide
- B. sulphur
- C. starch
- D. soap flakes

ANSWER: B

Discoloration should be avoided while handling

- A. silica flour
- B. tobacco
- C. zinc oxide
- D. starch

ANSWER: C

Which of the following characteristic can be assigned to soda ash?

- A. caustic
- B. fragile
- C. explosive
- D. free flowing

ANSWER: A

Chains for material handling equipment are generally made of

- A. cast iron
- B. wrought iron
- C. mild steel
- D. carbon steel

ANSWER: D

Which of the following cannot be handled using lifting magnets?

- A. Scrap iron
- B. mild steel plates
- C. high manganese steel
- D. castings of cast iron

ANSWER: C

Rails for railway track are made of

- A. cast iron
- B. mild steel
- C. chrome steel
- D. high speed steel

ANSWER: C

Which one of the following does not fall under the category of hoisting equipment?

- A. pull lift
- B. jack
- C. chain hoist
- D. dragline

ANSWER: D

Flight conveyors are mainly used for conveying

- A. grains
- B. coal

- C. bauxite
- D. iron ore

ANSWER: B

The number of cylinders in case of steam locomotive is

- A. One
- B. two
- C. four
- D. eight

ANSWER: B

In case of belt conveyors, the bearings used for return idlers are

- A. bush bearings
- B. split bush bearings
- C. cast iron bearings
- D. anti-friction bearings

ANSWER: D

Pneumatic conveyors are generally used for conveying

- A. package goods
- B. mineral ores
- C. heavy goods
- D. granular material

ANSWER: D

Which of the following bearing is not used in earth moving equipment?

- A. ball bearing
- B. bush bearing
- C. jewel bearing
- D. needle bearing

ANSWER: C

Lifting magnets cannot be used for

- A. under water handling
- B. handling materials above dull red heat
- C. handling dust of magnetic materials
- D. handling ferrous castings

ANSWER: B

Lift trucks are used in industries generally for the transportation of

- A. batches of material
- B. heavy equipment
- C. nonferrous materials
- D. castings only

ANSWER: A

Which of the following item can be conveyed using pneumatic conveyors?

- A. soda ash
- B. salt cake
- C. carbon black
- D. all of the above

ANSWER: D

Which braking system is not used in case of earth moving machinery?

- A. pneumatic braking system
- B. hydraulic braking system
- C. vacuum braking system
- D. mechanical braking system

ANSWER: C

Braking system is invariably incorporated in case of

- A. electric hoists
- B. magnetic hoists
- C. hand chain hoists
- D. all of the above

ANSWER: A

Escalators have the advantage of

- A. continuity of motion
- B. greater capacity
- C. small space requirement
- D. all of the above

ANSWER: D

Flight conveyors are used for the transportation of

- A. granular materials
- B. lumpy materials
- C. pulverized materials
- D. all of the above

ANSWER: D

Chains for material handling equipment are generally made of

- A. carbon steel
- B. cast iron
- C. wrought iron
- D. mild steel

ANSWER: A

Which one of the following is not a excavating and moving type machine?

- A. dump truck
- B. scarper
- C. clam shell
- D. dragline

ANSWER: A

Which of the following bearing is not used in earth moving equipment?

- A. jewel bearing
- B. needle bearing
- C. bush bearing
- D. ball bearing

ANSWER: A

Which of the following cannot be handled using lifting magnets?

- A. castings of cast iron
- B. high manganese steel
- C. mild steel plates
- D. scarp iron

ANSWER: B

In case of belt conveyors, the bearings used for return idlers are

- A. anti friction bearings
- B. bush bearings
- C. cast iron bearings
- D. split bush bearings

ANSWER: A

A four wheel drive has

- A. differential on front side only
- B. differential on rear side only
- C. both front as well as rear differentials
- D. no differential

ANSWER: C

In which of the following mechanism the relative motions of the rigid bodies are in one plane or in parallel planes?

- A. spatial mechanism
- B. spherical mechanism
- C. flexure mechanism
- D. planar mechanism

ANSWER: D

If there are redundant constraints in the chain and it forms a statically indeterminate structure,

what is the degree of freedom or the mobility of this structure?

- A. More than 0
- B. Less than 0
- C. 1
- D. 2

ANSWER: B

If the shortest link is fixed, what type of mechanism is obtained?

- A. Crank rocker mechanism
- B. Linkage is not planar
- C. Double crank mechanism
- D. Double rocker mechanism

ANSWER: C

Which of the following mechanism is an approximately straight-line motion mechanism?

- A. Hart's mechanism
- B. Watt's mechanism
- C. Scott Russell's mechanism
- D. Peaucellier mechanism

ANSWER: B

_____ Mechanism is a crossed four bar chain mechanism in early steam engines to guide the piston rod in a cylinder to have an approximate straight line motion.

- A. Chebychev's
- B. Watt's
- C. Peaucellier's
- D. Grasshoppe

ANSWER: B

Which of these is not a type of clutch?

- A. Cylindrical
- B. Centrifugal
- C. Single disc
- D. Conical

ANSWER: A

In a conical clutch, the mean radius of the bearing surface is 300 mm whereas the breadth is 20 mm. Find the inner and outer radii. The semi cone angle is 30° .

- A. 140 mm, 160 mm
- B. 160 mm, 140 mm
- C. 155 mm, 145 mm
- D. 145 mm, 155 mm

ANSWER: C

The smaller and generally the driving gear of a pair of mated gears is called _____

- A. Rack
- B. Pinion
- C. Module
- D. Pitch

ANSWER: B

The distance measured along the circumference of the pitch circle from a point on one tooth to the same point on the adjacent tooth is called _____

- A. circular pitch
- B. line of centers
- C. pitch diameter
- D. pitch point

ANSWER: A

The locus of a point on the pitch circle from the beginning to end of engagement of two mating gears is called as _____

- A. Arc of contact
- B. Path of contact
- C. Path of approach
- D. Arc of approach

ANSWER: A

The ratio of angle of action to the pitch angle is called _____

- A. angle of recess
- B. contact ratio
- C. space width
- D. angle of approach

ANSWER: B

The angle turned by the gear from the beginning to the end of engagement of a pair of teeth is called _____

- A. angle of action
- B. angle of contact
- C. angle of recess
- D. angle of approach

ANSWER: B

The condition which must be fulfilled by two gear tooth profiles to maintain a constant angular velocity ratio between them is called _____

- A. path of contact
- B. interference
- C. arc of contact
- D. law of gearing

ANSWER: D

The distance travelled by a point on either pitch circle of the two wheels during the period of contact of a pair of teeth is called _____

- A. angle of action
- B. path of contact
- C. arc of contact
- D. contact ratio

ANSWER: C

The contact ratio is the ratio of arc of contact to the _____

- A. Module
- B. circular pitch
- C. dedendum
- D. circular pitch

ANSWER: D

When large gear reductions are needed _____ gears are used.

- A. Bevel
- B. Worm
- C. Helical
- D. Spur

ANSWER: B

Which of the following is an advantage of worm gear?

- A. Has high power losses and low transmission efficiency
- B. Produce a lot of heat
- C. Used for reducing speed and increasing torque
- D. It is expensive

ANSWER: C

The distance by which a helix advances along the axis of the gear for one turnaround is called

- _____
- A. Lead
 - B. axial pitch
 - C. joint line
 - D. normal link

ANSWER: A

Bevel gears are equivalent to rolling _____

- A. Spheres
- B. Cubes
- C. Cuboids
- D. Cones

ANSWER: D

Which of the gears has the highest contact ratio?

- A. Worm
- B. Bevel
- C. Helical
- D. Spur

ANSWER: B

Which of the following clutches include shoes and spider inside the rim of the pulley?

- A. Cone clutch
- B. Multi plate clutch
- C. Single plate clutch
- D. Centrifugal clutch

ANSWER: D

In a multi plate clutch, number of discs on the driving shaft is given by $n-1$. True or false?

- A. True
- B. False

ANSWER: A

The quick return mechanism which is an inversion of 4-bar linkage is

- A. Drag link mechanism
- B. Whitworth quick return mechanism
- C. Crank and slotted lever mechanism
- D. None of the mentioned

ANSWER: A

When a cylinder is located in a Vee-block, the number of degrees of freedom which are arrested is

- A. 2
- B. 4
- C. 7
- D. 8

ANSWER: B

The number of degree of freedom of a planer linkage with 8 links and 9 simple revolute joints is

- A. 1
- B. 2
- C. 3
- D. 4

ANSWER: C

In a dynamically equivalent system, a uniformly distributed mass is divided into

- A. Three point masses
- B. Four point masses
- C. Two point masses
- D. Infinite point masses

ANSWER: C

The magnitude of velocities of the points on a rigid link is directly proportional to the distances from the points to the instantaneous center.

- A. True
- B. False

ANSWER: B

The velocity of the instantaneous center relative to any third rigid link will be different.

- A. True
- B. False

ANSWER: B

When the pin connects one sliding member and the other turning member, the angular velocity of the sliding member is _____

- A. 0
- B. 1
- C. 2
- D. 3

ANSWER: A

In a four stroke I.C. engine, the turning moment during the compression stroke is

- A. positive throughout
- B. negative throughout
- C. positive during major portion of the stroke
- D. negative during major portion of the stroke

ANSWER: B

The maximum fluctuation of energy is the

- A. difference between the maximum and minimum energies
- B. sum of the maximum and minimum energies
- C. variations of energy above and below the mean resisting torque to the
- D. ratio of the mean resisting torque to the work done per cycle

ANSWER: A

The co-efficient of fluctuation of energy is the ratio of maximum energy to the minimum energy.

- A. True
- B. False

ANSWER: B

The ratio of maximum fluctuation of energy to the work done per cycle is called

- A. fluctuation of energy
- B. maximum fluctuation of energy
- C. coefficient of fluctuation of speed
- D. none of the mentioned

ANSWER: C

What is the primary purpose of refrigeration and air-conditioning systems?

- A. To cool the air
- B. To heat the air
- C. To remove moisture from the air
- D. To regulate the temperature and humidity of a space

ANSWER: D

Which of the following is not an application of refrigeration and air-conditioning systems?

- A. Food preservation
- B. Comfort cooling
- C. Medical procedures
- D. Transportation

ANSWER: D

Which of the following refrigerants is not considered an ozone-depleting substance?

- A. R-22
- B. R-134a
- C. R-410A
- D. Ammonia

ANSWER: D

Which type of air-conditioning system is commonly used in residential homes?

- A. Split system
- B. Packaged system
- C. Central system
- D. All of the above

ANSWER: A

What is the purpose of an evaporator in a refrigeration system?

- A. To compress the refrigerant
- B. To remove heat from the system
- C. To expand the refrigerant
- D. To regulate the flow of refrigerant

ANSWER: B

Which of the following is an example of a commercial application of refrigeration?

- A. Home refrigerators
- B. Car air conditioning

- C. Ice cream freezers
- D. Window air conditioners

ANSWER: C

Which type of air conditioning system is commonly used in large commercial buildings?

- A. Split system
- B. Packaged system
- C. Central system
- D. Window unit

ANSWER: C

Which of the following is a common application of air conditioning in the healthcare industry?

- A. Patient comfort
- B. Equipment cooling
- C. Sterilization
- D. All of the above

ANSWER: D

What is the most commonly used cryogenic fluid?

- A. Liquid helium
- B. Liquid nitrogen
- C. Liquid oxygen
- D. Liquid carbon dioxide

ANSWER: B

What is the primary use of cryogenics?

- A. Food preservation
- B. Medical research
- C. Space exploration
- D. All of the above

ANSWER: D

What is a cry cooler?

- A. A device used to cool objects to very low temperatures
- B. A device used to heat objects to very high temperatures
- C. A device used to generate electricity from cryogenic fluids
- D. A device used to measure the temperature of cryogenic fluids

ANSWER: A

What safety precautions should be taken when working with cryogenic fluids?

- A. Wear appropriate personal protective equipment, such as goggles and gloves
- B. Store and handle cryogenic fluids in a well-ventilated area
- C. Post appropriate warning signs
- D. All of the above

ANSWER: D

What is the purpose of cryopreservation?

- A. To freeze and store biological materials at very low temperatures to preserve them for future use
- B. To liquefy cryogenic fluids for transportation
- C. To generate electricity from cryogenic fluids
- D. To measure the temperature of cryogenic fluids

ANSWER: A

Which of the following is an example of a cryogenic application?

- A. Air conditioning in a residential home
- B. Cooling of computer processors
- C. Refrigeration of food products
- D. Cooling of medical equipment during surgeries

ANSWER: D

What is a liquefied natural gas (LNG) carrier?

- A. A refrigeration system used to transport natural gas at low temperatures
- B. A refrigeration system used to transport liquid nitrogen at low temperatures
- C. A refrigeration system used to transport liquid oxygen at low temperatures
- D. A refrigeration system used to transport liquid helium at low temperatures

ANSWER: A

What is a cool room?

- A. A room used for storing perishable items at low temperatures
- B. A room designed to maintain a cool temperature for comfort
- C. A room used for storing electronics at low temperatures
- D. A room used to simulate low temperature environments for research purposes

ANSWER: A

What is a refrigerated container?

- A. A container used to transport perishable items at low temperatures
- B. A container used to transport electronics at low temperatures
- C. A container used to store liquid nitrogen at low temperatures
- D. A container used to store medical equipment at low temperatures

ANSWER: A

What is a clean room?

- A. A room that has been cleaned with special cleaning agents
- B. A room designed to maintain a specific level of cleanliness and control of environmental factors
- C. A room that has been treated with ultraviolet light to reduce bacteria and viruses
- D. A room that has a high level of negative pressure to prevent the spread of contaminants

ANSWER: B

What is a chilled beam system?

- A. A system that uses a fan to blow air over a cooled surface to provide cooling
- B. A system that uses a combination of air and water to provide cooling
- C. A system that uses an evaporative cooling process to provide cooling
- D. A system that uses a refrigerant to provide cooling directly to the air stream

ANSWER: B

Which of the following is a common application of air conditioning in data centers?

- A. Comfort cooling for employees
- B. Cooling of computer processors
- C. Humidity control to prevent equipment damage
- D. All of the above

ANSWER: B

What is a desiccant dehumidification system?

- A. A system that uses a refrigerant to cool and dehumidify the air
- B. A system that uses a desiccant material to absorb moisture from the air
- C. A system that uses a chemical process to remove moisture from the air
- D. A system that uses a fan to blow air over a cooled surface to remove moisture

ANSWER: B

What is a variable refrigerant flow (VRF) system?

- A. A system that can vary the amount of refrigerant flow to different indoor units based on their cooling load
- B. A system that uses a variable speed compressor to adjust the cooling capacity
- C. A system that uses a variable speed fan to adjust the airflow
- D. A system that uses a variable moisture control mechanism to regulate humidity

ANSWER: A

What is the working fluid used in an air cycle refrigeration system?

- A. Water
- B. Ammonia
- C. Carbon dioxide
- D. Air

ANSWER: D

Which of the following is not a component of an air cycle refrigeration system?

- A. Compressor
- B. Condenser
- C. Evaporator
- D. Expansion valve

ANSWER: B

How does an air cycle refrigeration system work?

- A. It compresses air, cools it, and expands it to produce cooling
- B. It compresses refrigerant, cools it, and expands it to produce cooling
- C. It evaporates water, cools the air, and condenses the water to produce cooling
- D. It boils a liquid, cools the vapor, and condenses it to produce cooling

ANSWER: A

What is the primary application of air cycle refrigeration systems?

- A. Comfort cooling in residential homes
- B. Cooling of electronic equipment
- C. Cooling of aircraft cabins
- D. Cooling of food products

ANSWER: C

Which of the following is not an advantage of air cycle refrigeration systems?

- A. They have low environmental impact
- B. They are highly efficient
- C. They use a non-toxic working fluid
- D. They are easy to maintain

ANSWER: B

What is a heat exchanger in an air cycle refrigeration system?

- A. A device used to transfer heat from the compressed air to the surrounding environment
- B. A device used to transfer heat from the evaporated working fluid to the compressed air
- C. A device used to transfer heat from the compressed air to the evaporated working fluid
- D. A device used to transfer heat from the condenser to the evaporator

ANSWER: C

Which of the following is not a limitation of air cycle refrigeration systems?

- A. They have a low coefficient of performance (COP)
- B. They require a large amount of space to operate
- C. They are not effective at very low temperatures
- D. They are not effective at very high temperatures

ANSWER: B

What is a bleed air system in an air cycle refrigeration system?

- A. A system that uses air from the compressor to provide cooling
- B. A system that uses air from the cabin to provide cooling
- C. A system that uses air from the engine to provide cooling
- D. A system that uses air from the outside environment to provide cooling

ANSWER: C

What is a ram air system in an air cycle refrigeration system?

- A. A system that uses air from the cabin to provide cooling
- B. A system that uses air from the outside environment to provide cooling
- C. A system that uses air from the compressor to provide cooling
- D. A system that uses air from the engine to provide cooling

ANSWER: B

What is the purpose of the expansion valve in an air cycle refrigeration system?

- A. To reduce the pressure and temperature of the working fluid
- B. To increase the pressure and temperature of the working fluid
- C. To regulate the flow of the working fluid
- D. To transfer heat from the compressed air to the evaporated working fluid

ANSWER: A

What is the primary disadvantage of air cycle refrigeration systems compared to vapor compression refrigeration systems?

- A. They are less reliable
- B. They are less efficient
- C. They require more maintenance
- D. They are more expensive

ANSWER: B

What is the purpose of the compressor in an air cycle refrigeration system?

- A. To compress the working fluid and increase its pressure and temperature
- B. To transfer heat from the compressed air to the evaporated working fluid
- C. To regulate the flow of the working fluid
- D. To reduce the pressure and temperature of the working fluid

ANSWER: A

What is the main assumption made in air standard analysis?

- A. The working fluid is a perfect gas
- B. The working fluid is a liquid
- C. The working fluid is a mixture of gases
- D. The working fluid is a solid

ANSWER: A

In the reversed Carnot cycle, which process is reversible adiabatic compression or expansion?

- A. Compression
- B. Expansion
- C. Both are reversible adiabatic processes
- D. Neither is a reversible adiabatic process

ANSWER: C

What is the efficiency of the reversed Carnot cycle?

- A. Greater than or equal to the efficiency of any other cycle operating between the same

temperature limits

- B. Less than the efficiency of any other cycle operating between the same temperature limits
- C. Equal to the efficiency of a Carnot cycle operating between the same temperature limits
- D. Equal to the efficiency of a Rankine cycle operating between the same temperature limits

ANSWER: A

In the reversed Carnot cycle, what is the state of the gas at the end of the isothermal expansion process?

- A. Saturated liquid
- B. Saturated vapor
- C. Superheated vapor
- D. Sub cooled liquid

ANSWER: C

In the reversed Carnot cycle, what is the purpose of the isothermal expansion process?

- A. To remove heat from the working fluid
- B. To increase the temperature and pressure of the working fluid
- C. To decrease the temperature and pressure of the working fluid
- D. To maintain the temperature and pressure of the working fluid

ANSWER: A

What is the refrigerant in a typical vapor compression refrigeration system?

- A. Water
- B. Air
- C. CO₂
- D. A refrigerant such as R-134a or R-22

ANSWER: D

Which component of a vapor compression refrigeration system is responsible for metering the flow of refrigerant into the evaporator?

- A. Compressor
- B. Condenser
- C. Expansion valve
- D. Evaporator

ANSWER: C

What is the coefficient of performance (COP) of a vapor compression refrigeration system?

- A. The ratio of the heat removed to the power input
- B. The ratio of the heat removed to the heat added
- C. The ratio of the power input to the heat added
- D. The ratio of the heat added to the heat removed

ANSWER: A

Which law of thermodynamics describes the basic principle behind the operation of a vapor

compression refrigeration system?

- A. First law of thermodynamics
- B. Second law of thermodynamics
- C. Third law of thermodynamics
- D. Zeros law of thermodynamics

ANSWER: B

Which of the following is a key difference between gas cycles and vapor cycles?

- A. Gas cycles use a working fluid that is in a gaseous state throughout the cycle, while vapor cycles use a working fluid that undergoes a phase change during the cycle.
- B. Gas cycles are used primarily for heating applications, while vapor cycles are used primarily for cooling applications.
- C. Gas cycles are more efficient than vapor cycles.
- D. Vapor cycles operate at higher pressures than gas cycles.

ANSWER: A

In a vapor compression refrigeration cycle, which component is responsible for changing the refrigerant from a high-pressure vapor to a low-pressure vapor?

- A. Compressor
- B. Condenser
- C. Expansion valve
- D. Evaporator

ANSWER: C

In a standard vapor compression refrigeration system, which component is responsible for removing heat from the cooled space?

- A. Compressor
- B. Condenser
- C. Expansion valve
- D. Evaporator

ANSWER: D

What is the effect of increasing the condenser temperature in a standard vapor compression refrigeration system?

- A. It increases the COP of the system
- B. It decreases the COP of the system
- C. It has no effect on the COP of the system
- D. It increases the power input of the system

ANSWER: B

In a vapor absorption refrigeration system, which component is responsible for absorbing the refrigerant vapor and returning it to a liquid state?

- A. Evaporator
- B. Condenser

- C. Absorber
 - D. Generator
- ANSWER: C

What is the function of the generator in a vapor absorption refrigeration system?

- A. To cool the refrigerant vapor and condense it into a liquid
- B. To evaporate the refrigerant
- C. To regulate the flow of refrigerant
- D. To separate the refrigerant from the absorbent and raise its pressure

ANSWER: D

What is the coefficient of performance (COP) of a vapor absorption refrigeration system?

- A. The ratio of the heat removed to the power input
- B. The ratio of the heat removed to the heat added
- C. The ratio of the power input to the heat added
- D. The ratio of the heat added to the heat removed

ANSWER: B

In a vapor absorption refrigeration system based on the ammonia-water pair, what is the role of water?

- A. Water is used as the refrigerant.
- B. Water is used as the absorbent.
- C. Water is used as the working fluid in the generator.
- D. Water is used as the working fluid in the condenser.

ANSWER: B

In a vapor absorption refrigeration system based on the ammonia-water pair, what is the role of ammonia?

- A. Ammonia is used as the refrigerant.
- B. Ammonia is used as the absorbent.
- C. Ammonia is used as the working fluid in the generator.
- D. Ammonia is used as the working fluid in the condenser.

ANSWER: A

In a vapor absorption refrigeration system based on the ammonia-water pair, what is the purpose of the evaporator?

- A. The evaporator is where the refrigerant is condensed.
- B. The evaporator is where the refrigerant is absorbed by the absorbent.
- C. The evaporator is where the refrigerant is compressed.
- D. The evaporator is where the cooling effect is produced.

ANSWER: D

Which of the following factors is NOT a consideration when selecting a refrigerant for a system?

- A. Cost

- B. Environmental impact
- C. Compressor power input
- D. Compatibility with materials

ANSWER: C

What is the term for the ratio of the mass of water vapor in a given volume of air to the total mass of the air?

- A. Humidity ratio
- B. Relative humidity
- C. Dew point temperature
- D. Wet bulb temperature

ANSWER: A

What is the term for the temperature at which the air becomes saturated with water vapor and condensation begins to form?

- A. Humidity ratio
- B. Relative humidity
- C. Dew point temperature
- D. Wet bulb temperature

ANSWER: C

Which of the following processes involves adding moisture to air without changing its temperature?

- A. Sensible heating
- B. Sensible cooling
- C. Adiabatic cooling
- D. Humidification

ANSWER: D

Which of the following parameters is a measure of the moisture content of air?

- A. Wet bulb temperature
- B. Dry bulb temperature
- C. Relative humidity
- D. Enthalpy

ANSWER: C

What is the term for the temperature at which the air would need to be cooled in order for it to become saturated with water vapor?

- A. Wet bulb temperature
- B. Dry bulb temperature
- C. Dew point temperature
- D. Enthalpy

ANSWER: A

Which of the following processes involves cooling air without changing its moisture content?

- A. Sensible cooling
- B. Latent cooling
- C. Humidification
- D. Dehumidification

ANSWER: A

What is the definition of relative humidity?

- A. The amount of moisture in the air
- B. The ratio of actual vapor pressure to the saturation vapor pressure at a given temperature
- C. The ratio of dry air to water vapor in a mixture
- D. The temperature at which water vapor begins to condense out of the air

ANSWER: B

What is the inside design condition for a typical air conditioning system in a commercial building?

- A. 20°C dry bulb temperature, 50% relative humidity
- B. 23°C dry bulb temperature, 50% relative humidity
- C. 25°C dry bulb temperature, 50% relative humidity
- D. 27°C dry bulb temperature, 50% relative humidity

ANSWER: C

What is the formula for calculating the total cooling load of a space?

- A. Total cooling load = Sensible cooling load + Latent cooling load
- B. Total cooling load = Sensible cooling load - Latent cooling load
- C. Total cooling load = Sensible cooling load $\times$ Latent cooling load
- D. Total cooling load = Sensible cooling load $\div$ Latent cooling load

ANSWER: A

Which of the following air conditioning systems is most suitable for large commercial spaces?

- A. Window air conditioning units
- B. Split air conditioning systems
- C. Central air conditioning systems
- D. Portable air conditioning units

ANSWER: C

Which of the following factors is important to consider when selecting an air conditioning system for a residential space?

- A. Noise level of the system
- B. Capacity of the system
- C. Type of refrigerant used
- D. Cost of the system

ANSWER: A

Which of the following air conditioning systems is a type of rooftop system?

- A. Window air conditioning units
- B. Split air conditioning systems
- C. Packaged air conditioning systems
- D. Portable air conditioning units

ANSWER: C

Which of the following factors is important in determining the efficiency of an air conditioning system?

- A. Size of the space to be cooled
- B. Type of refrigerant used
- C. Color of the walls in the space
- D. Number of windows in the space

ANSWER: B

What is the main purpose of air conditioning ducts?

- A. To distribute air throughout a space
- B. To filter air before it enters a space
- C. To heat air before it enters a space
- D. To cool air before it enters a space

ANSWER: A

What is the most common material used for air conditioning ducts?

- A. Steel
- B. Copper
- C. Aluminum
- D. PVC

ANSWER: C

What is the purpose of insulation in air conditioning ducts?

- A. To prevent air leakage
- B. To reduce noise transmission
- C. To improve energy efficiency
- D. All of the above

ANSWER: D

What is the term used to describe the resistance of air flow in ductwork?

- A. Velocity pressure
- B. Static pressure
- C. Total pressure
- D. Dynamic pressure

ANSWER: B

Which of the following factors can affect the transmission of air in air conditioning ducts?

- A. Duct size and shape
- B. Air velocity
- C. Type of duct material
- D. All of the above

ANSWER: D

Which of the following duct shapes has the lowest pressure drop?

- A. Round ducts
- B. Rectangular ducts
- C. Oval ducts
- D. Triangular ducts

ANSWER: A

What is the main purpose of air ducts in an air conditioning system?

- A. To filter the air
- B. To humidify the air
- C. To transport the air
- D. To cool the air

ANSWER: C

What is the purpose of a damper in an air conditioning duct?

- A. To regulate the flow of air
- B. To filter the air
- C. To insulate the ducts
- D. To humidify the air

ANSWER: A

Which of the following types of fans is commonly used in air conditioning ducts?

- A. Centrifugal fans
- B. Axial fans
- C. Pedestal fans
- D. Ceiling fans

ANSWER: A

in the evaporator of a vapor compression refrigeration system:

- A. A low temperature is maintained so that heat can flow from the external fluid
- B. Refrigeration effect is produced as the refrigerant liquid vaporizes
- C. A low pressure is maintained so that the compressor can run
- D. All except c

ANSWER: D

The function of a compressor in a vapor compression refrigeration system is to:

- A. To maintain the required low-side pressure in the evaporator
- B. To maintain the required high-side pressure in the condenser

- C. To circulate required amount of refrigerant through the system
- D. all of the above

ANSWER: D

In an air cycle refrigeration system, low temperatures are produced due to:

- A. Evaporation of liquid air
- B. Throttling of air
- C. Expansion of air in turbine
- D. None of the above

ANSWER: C

Absorption of the refrigerant by the absorbent in vapor absorption refrigeration System is accompanied by:

- A. Absorption of heat
- B. Release of heat
- C. No thermal effects
- D. Reduction in volume

ANSWER: B

Which of the following statements are TRUE?

- A. During sensible cooling of air, both dry bulb and wet bulb temperatures decrease
- B. During sensible cooling of air, dry bulb temperature decreases but wet bulb temperature remains constant
- C. During sensible cooling of air, dry and wet bulb temperatures decrease but dew point temperature remains constant
- D. A and C

ANSWER: D

Which of the following statements are TRUE?

- A. During cooling and humidification process, the enthalpy of air decreases
- B. During cooling and humidification process, the enthalpy of air increases
- C. During cooling and humidification process, the enthalpy of air remains constant
- D. During cooling and humidification process, the enthalpy of air may increase, decrease or remain constant depending upon the temperature of the wet surface

ANSWER: D

An air stream at a flow rate of 1 kg/s and a DBT of 30°C mixes adiabatically with another air stream flowing with a mass flow rate of 2 kg/s and at a DBT of 15°C. Assuming no condensation to take place, the temperature of the mixture is approximately equal to:

- A. 20°C
- B. 22.5°C
- C. 25°C
- D. Cannot be found

ANSWER: A

What is the required wattage of an electrical heater that heats 0.1 m³/s of air from 15°C and 80% RH to 55°C? The barometric pressure is 101.325 kPa. Given that average specific heat of air ($C_{pm}=1.021\text{kJ/kg.k}$), vapor pressure ($P_v=1.364\text{kPa}$) and gas constant of air (0.287kJ/kg.k).

- A. 6.3kW
- B. 4.938kW
- C. 5.8kW
- D. 13kW

ANSWER: B

Within elastic limit in a loaded material, stress is

- A. inversely proportional to strain
- B. directly proportional to strain
- C. equal to strain
- D. none of the above.

ANSWER: B

The ratio of linear stress to linear strain is known as

- A. Poisson's ratio
- B. bulk modulus
- C. modulus of rigidity
- D. modulus of elasticity.

ANSWER: D

The ratio of lateral strain to longitudinal strain is called

- A. Poisson's ratio
- B. bulk modulus
- C. modulus of rigidity
- D. modulus of elasticity.

ANSWER: A

The ratio of shear stress to shear strain is called

- A. Poisson's ratio
- B. bulk modulus
- C. modulus of rigidity
- D. modulus of elasticity.

ANSWER: C

The ratio of normal stress of each face of a solid cube to volumetric strain is called

- A. Poisson's ratio
- B. bulk modulus
- C. modulus of rigidity
- D. modulus of elasticity.

ANSWER: B

Hooke's law holds good upto

- A. proportional limit
- B. yield point
- C. elastic limit
- D. plastic limit.

ANSWER: A

The property of a material by virtue of which a body returns to its original shape after removal of the load is known as

- A. ductility
- B. plasticity
- C. elasticity
- D. resilience.

ANSWER: C

A tensile force (P) is acting on a body of length (L) and area of cross-section (A). The change in length would be

- A. $P \cdot L / AE$
- B. PE / LA
- C. PL / AE
- D. AL / PE

ANSWER: C

The modulus of elasticity (E) and modulus of rigidity (C) are related by

- A. $C = mE / (3(m-2))$
- B. $C = mE / (2(m+1))$
- C. $C = (3(m-2)) / mE$
- D. $C = (2(m+1)) / mE$ where, $1/m$ is Poisson's ratio

ANSWER: B

The modulus of elasticity (E) and bulk modulus (K) are related by

- A. $K = mE / (3(m-2))$
- B. $K = mE / (2(m+1))$
- C. $K = (3(m-2)) / mE$

(d) $K = \frac{2(m+1)}{mE}$ where, $\frac{1}{m}$ is Poisson's ratio

ANSWER: A

The elongation produced in a rod (by its own weight) of length (l) and diameter (d) rigidly fixed at the upper end and hanging is equal to

$\frac{\rho l^2}{2E}$

$\frac{\rho l^2}{2E}$

$\frac{\rho l^3}{2E}$

(d) $\frac{\rho l^4}{2E}$, where: ρ -weight per unit volume of the rod, E- is modulus of elasticity

ANSWER: B

The ratio of modulus of rigidity to modulus of elasticity for a Poisson's ratio of 0.25 would be

A. 0.5

B. 0.4

C. 0.3

D. 1.0.

ANSWER: B

The ratio of bulk modulus to modulus of elasticity for a Poisson's ratio of 0.25 would be

A. $\frac{2}{3}$

B. $\frac{1}{3}$

C. $\frac{4}{3}$

D. 1.0.

ANSWER: A

The relation between modulus of elasticity (E), modulus of rigidity (C) and bulk modulus (K) is given by

A. $E = \frac{3KC}{C+9K}$

B. $E = \frac{9KC}{C+3K}$

C. $E = \frac{C+9K}{3KC}$

D. $E = \frac{C+3K}{9KC}$

ANSWER: B

The ratio of modulus of rigidity to bulk modulus for a Poisson's ratio of 0.25 would be

A. $\frac{2}{3}$

B. $\frac{2}{5}$

C. $\frac{3}{5}$

D. 1.0.

ANSWER: C

The work done in producing strain on a material per unit volume is called

A. resilience

B. ductility

C. elasticity

D. plasticity

ANSWER: A

The property by virtue of which a metal can be beaten into plates is called

A. ductility

- B. malleability
- C. resilience
- D. plasticity.

ANSWER: B

Choose the wrong statement

- A. Elongation produced in a rod (by its own weight) which is rigidly fixed at the upper end and hanging is equal to that produced by a load half its weight applied at the end.
- B. The stress at any section of a rod on account of its own weight is directly proportional to the distance of the section from the lower end.
- C. Modulus of elasticity is having the same unit as stress.
- D. If a material expands freely due to heating, it will develop thermal stresses.

ANSWER: D

A rod of length (l) tapers uniformly from a diameter D1 to a diameter D2 and carries an axial tensile load P. The extension of the rod would be

- $\frac{PL}{4ED_1 D_2}$
- $\frac{4PL}{ED_1 D_2}$
- (c) $\frac{PEL}{4D_1 D_2}$
- $\frac{4PEL}{D_1 D_2}$

ANSWER: B

If in question 19, the rod is of uniform diameter (D), then extension would be

- $\frac{PL}{4ED^2}$
- $\frac{4PL}{ED^2}$
- (c) $\frac{PEL}{4D^2}$
- $\frac{4PEL}{D^2}$

ANSWER: B

For the bars of composite section,

the load carried by different materials is the same as the total external load

the total external load is equal to the total sum of the loads carried by different materials

strain in all materials is equal

both (c) and (d).

ANSWER: D

Two materials are having modulus of elasticities, modulus of rigidities and bulk modulus as (E1, E2); (C1, C2) and (K1, K2). The modulus ratio is given by

- $\frac{E_1}{C_2}$
- $\frac{E_1}{K_2}$
- (c) $\frac{E_1}{E_2}$
- $\frac{C_1}{K_2}$

ANSWER: C

The thermal stress is given by

- $E\alpha T$
- $E\alpha/T$
- (c) $E\alpha/T$
- $\frac{1}{E\alpha T}$

(d) $\frac{1}{E\alpha T}$ where α = co-efficient of linear expansion, and T = rise in temperature.

ANSWER: A

The proof resilience is given by

$(\sigma^2)/E$

$2E/\sigma^2$

(c) $\sigma^2/2E$

(d) $E/\sigma^2\sigma^2$ where σ = stress at the elastic limit.

ANSWER: C

If l be the length and D be the diameter of a cylindrical rod, then volumetric strain of the rod is equal to

strain of length plus strain of diameter

strain of diameter

strain of length + twice the strain of diameter

strain of length.

ANSWER: C

The extension per unit length of the rod due to suddenly applied load as compared to the same load gradually applied to the same rod is

same

double

three times

half.

ANSWER: B

The stress due to suddenly applied load as compared to the stress due to the same load gradually applied to the same rod is

half

same

double

three times.

ANSWER: C

The normal stress on an oblique plane at an angle θ to the cross-section of a body which is subjected to a direct tensile stress (σ) is equal to

$\sigma/2 \sin^2\theta$

$\sigma \cos^2\theta$

(c) $\sigma \cos^2\theta$

$\sigma \sin^2\theta$

ANSWER: C

In question 28, the normal stress on the oblique plane will be maximum when θ is equal to

90°

45°

0°

30°.

ANSWER: C

The tangential or shear stress on an oblique plane at an angle θ to the cross-section of a body which is subjected to a direct tensile stress (σ) is equal to

$$\frac{\sigma}{2} \sin 2\theta$$

$$\sigma \cos \theta$$

$$\left(\frac{\sigma}{2} \cos^2 \theta + \frac{\sigma}{2} \sin^2 \theta \right)$$

$$\frac{\sigma}{2} \sin^2 \theta$$

ANSWER: A

In question 30, the tangential or shear stress on the oblique plane will be maximum when θ is equal to

90°

45°

0°

30°.

ANSWER: B

The resultant stress on an oblique plane at an angle θ to the cross-section of a body which is subjected to a direct tensile stress is

$$\frac{\sigma}{2} \sin 2\theta$$

$$\sigma \cos \theta$$

$$\left(\frac{\sigma}{2} \cos^2 \theta + \frac{\sigma}{2} \sin^2 \theta \right)$$

$$\frac{\sigma}{2} \sin^2 \theta$$

ANSWER: B

If a member is subjected to an axial tensile load, the plane normal to the axis of loading carries
minimum normal stress
maximum normal stress
maximum shear stress
none of the above.

ANSWER: B

If a member is subjected to an axial tensile load, the plane inclined at 45° to the axis of loading carries

minimum shear stress

maximum normal stress

maximum shear stress

none of the above.

ANSWER: C

The maximum shear stress induced in a member which is subjected to an axial load is equal to
maximum normal stress

half of maximum normal stress
twice the maximum normal stress
thrice the maximum normal stress.

ANSWER: B

If a member, whose tensile strength is more than two times the shear strength, is subjected to an axial load upto failure, the failure of the member will occur by
maximum normal stress
maximum shear stress
normal stress or shear stress
none of the above.

ANSWER: B

If a beam is fixed at both its ends, it is called a
fixed beam
built-in beam
encastered beam
any one of the above

ANSWER: D

If a beam is supported on more than two supports, it is called a
built-in beam
continuous beam
simply supported beam
encastered beam.

ANSWER: B

Choose the wrong statement

The shear force at any section of a beam is equal to the total sum of the forces acting on the beam on any one side of the section.

The magnitude of the bending moment at any section of a beam is equal to the vector sum of the moments (about the section) due to the forces acting on the beam on any one side of the section.

A diagram which shows the values of shear forces at various sections of structural member is called a shear force diagram.

A simply supported beam is one which is supported on more than two supports.

ANSWER: D

A simply supported beam of span (l) carries a point load (W) at the center of the beam. The bending moment diagram will be a
parabola with maximum ordinate at the center of the beam
parabola with maximum ordinate at one end of the beam
triangle with maximum ordinate at the center of the beam
triangle with maximum ordinate at one end of the beam.

ANSWER: C

A simply supported beam of span (l) carries a uniformly distributed load (w N per unit length) over the whole span. The bending moment diagram will be a
parabola with maximum ordinate at the center of the beam
parabola with maximum ordinate at one end of the beam
triangle with maximum ordinate at the center of the span
triangle with maximum ordinate at one end of the beam.

ANSWER: A

A cantilever of length (l) carries a point load (W) at the free end. The bending moment diagram will be a
parabola with maximum ordinate at the center of the beam
parabola with maximum ordinate at the cantilever end
triangle with maximum ordinate at the free end
triangle with maximum ordinate at the cantilever end.

ANSWER: D

A cantilever of length (l) carries a uniformly distributed load over the whole length. The bending moment diagram will be
parabola with maximum ordinate at the center
parabola with maximum ordinate at the cantilever end
triangle with maximum ordinate at the free end
triangle with maximum ordinate at the cantilever end.

ANSWER: B

A simply supported beam of span (l) carries a point load (W) at the center of the beam. The shear force diagram will be
a rectangle
a triangle
two equal and opposite rectangles
two equal and opposite triangles.

ANSWER: C

A simply supported beam of span (l) carries a uniformly distributed load over the whole span. The shear force diagram will be
a rectangle
a triangle
two equal and opposite rectangles
two equal and opposite triangles.

ANSWER: D

A cantilever of length (l) carries a point load (W) at the free end. The shear force diagram will be
two equal and opposite rectangles

a rectangle
two equal and opposite triangles
a triangle.

ANSWER: B

A cantilever of length (l) carries a uniformly distributed load over the whole length. The shear force diagram will be
two equal and opposite rectangles
a rectangle
two equal and opposite triangles
a triangle.

ANSWER: B

The bending moment on a section is maximum where shearing force is
minimum
maximum
zero
changing sign

ANSWER: D

The point of zero bending moment, where the continuous curve of bending moment changes sign, is called
the point of contraflexure
the point of inflation
the point of virtual hinge
all of the above.

ANSWER: D

The point of contraflexure occurs only in
continuous beams
cantilever beams
overhanging beams
simply supported beams

ANSWER: C

The bending moment at a section, where shear force is zero, will be
zero
maximum
minimum
either minimum or maximum

ANSWER: D

A beam of uniform strength is one which has same
bending stress at every section
deflection throughout the beam

bending moment throughout the beam
shear force throughout the beam.

ANSWER: A

The shear force and bending moment are zero at the free end of a cantilever, if it carries a
point load at the free end
uniformly distributed load over the whole length
point load in the middle of its length
none of the above.

ANSWER: B

A simply supported beam carries a uniformly distributed load of w N per unit length over the whole span (l). The shear force at the center is

$wl/2$

$(wl^2)/8$

$wl/4$

zero

ANSWER: D

For the question 54, the shear force at the supported ends will be

$\pm wl/2$

$(wl^2)/8$

$wl/4$

zero

ANSWER: A

In a tensile test of a specimen, the ratio of maximum load to the original cross-sectional area of the test piece is called

yield stress

ultimate stress

safe stress

breaking stress.

ANSWER: B

A short column of rectangular section carries a vertical point load W axially. The stress on the section of the column will be

uniform

zero at one end and maximum at the other end

zero at the axis and maximum on the outer ends

tensile on one end and compressive on the other.

ANSWER: A

If in question 57, the short column carries a vertical point load eccentrically, then the stress on the section will be

uniform

zero at the axis and maximum on the outer ends

zero at one end and maximum at the other end

non-uniform and will depend upon the amount of eccentricity.

ANSWER: D

A solid circular shaft of diameter D carries an axial load W . If the same load is applied axially on a hollow circular shaft of inner diameter as $D/2$, the ratio of stresses in a solid shaft to that of hollow shaft would be

$1/2$

$1/4$

$4/3$

$3/4$

ANSWER: D

The diameter of a mild steel round bar, on which tensile test is performed, at fracture will increase

decrease

be same

none of the above.

ANSWER: B

If a member is subjected to a uniform bending moment (M), the radius of curvature of the deflected form of the member is given by

$M/R = E/I$

$M/I = E/R$

$M/I = R/E$

$M/E = RI$

ANSWER: B

The expression $EI (d^2 y)/(dx^2)$ at a section of a member represents

shearing force

rate of loading

bending moment

slope.

ANSWER: C

The expression $EI (d^3 y)/(dx^3)$ at a section of a member represents

shearing force

rate of loading

bending moment

slope.

ANSWER: A

The expression $EI (d^4 y)/(dx^4)$ at a section of a member represents
shearing force
rate of loading
bending moment
slope.

ANSWER: B

The statement that 'the deflection at any point in a beam subjected to any load system is equal to the partial derivative of the total strain energy stored with respect to the load acting at the point in the direction in which deflection is desired' is called

Bettles's law
the first theorem of Castigliano
Clapeyron's theorem
Maxwell's law.

ANSWER: B

A laminated spring 1 m long carries a central point load of 2000 N. The spring is made up of plates each 5 cm wide and 1 cm thick. The bending stress in the plates is limited to 10 N/mm². The number of plates required, will be

3
5
6
8.

ANSWER: C

In question 66, if $E = 2 \times 10^5$ N/mm² the deflection under the given load of 2000 N will be

1 cm
1.25 cm
1.3 cm
1.40 cm.

ANSWER: B

A fixed beam is a beam whose end supports are such that the end slopes
are maximum
are minimum
are zero
none of the above.

ANSWER: C

A fixed beam of length (l) carries a point load (W) at the center. The deflection at the center is
same as for a simply supported beam
half of the deflection for a simply supported beam
one-fourth of the deflection for a simply supported beam
Double the deflection for a simply supported beam.

ANSWER: C

For the question 69, the number of points of contra flexure

is one
are two
are three
is none.

ANSWER: B

For the question 69, the points of contra flexure lies at
the fixed ends
the middle of the beam
 $l/4$ from the ends
None of the above.

ANSWER: C

For the question 69, the bending moment at the center is
same as for a simply supported beam
half of the bending moment for a simply supported beam
one-fourth of the bending moment for a simply supported beam
Double the bending moment for a simply supported beam.

ANSWER: B

The ratio of moment of inertia about the neutral axis to the distance of the most distance point of the section from the neutral axis is called
moment of inertia
section modulus
polar moment of inertia
Modulus of rigidity.

ANSWER: B

Fletched beam means a
continuous beam
fixed beam
beam of composite section consisting of a wooden beam strengthened by mild steel plates
None of the above.

ANSWER: C

The shear stress required to cause plastic deformation of solid metal is called
proof stress
flow stress
rupture stress
Ultimate stress.

ANSWER: B

The stress which will cause a specified permanent deformation in a material (usually 0.01% or less) is called
proof stress
flow stress
rupture stress
Ultimate stress.

ANSWER: A

The stress obtained by dividing the load at the moment of incipient fracture, by the area

supporting that load is called

proof stress

flow stress

rupture stress

Ultimate stress.

ANSWER: C

The statement that 'If unit loads rest upon a beam at the two points A and B, then the deflection at A due to unit load at B is equal to the deflection at B due to unit load at A' is given by

Mohr

Castigliano

Maxwell

Rankine.

ANSWER: C

A short column of rectangular section carries a point load (W) acting with an eccentricity (e). The shape of Kernel area would be

square

rectangle

circle

Rhombus.

ANSWER: D

If in question 79, the column is of circular section then the shape of Kernel area would be

square

rectangle

circle

Rhombus.

ANSWER: C

Every cross-section of a shaft, which is subjected to a twisting moment, is under compressive stress

shear stress

tensile stress

Bending stress.

ANSWER: B

The shear stress at any point of a shaft, subjected to twisting moment, is proportional to its distance from the central axis of the shaft

inversely proportional to its distance from the central axis of the shaft

proportional to the square of its distance from the central axis of the shaft

None of the above.

ANSWER: A

A solid shaft of diameter D transmits the torque equal to

$\frac{\pi}{32} D^3$

(b) $\frac{\pi}{64} D^3$

(c) $\frac{\pi}{16} D^3$

(d) $\frac{\pi}{8} D^3$

ANSWER: C

Torsional rigidity of a shaft is equal to
product of modulus of rigidity and polar moment of inertia
sum of modulus of rigidity and polar moment of inertia
difference of modulus of rigidity and polar moment of inertia
Ratio of modulus of rigidity and polar moment of inertia.

ANSWER: A

Polar modulus of a shaft section is equal to
product of polar moment of inertia and maximum radius of the shaft
ratio of polar moment of inertia to maximum radius of the shaft
sum of polar moment of inertia and maximum radius of the shaft
Difference of polar moment of inertia and maximum radius of the shaft.

ANSWER: B

The torsional rigidity of a shaft is defined as the torque required to produce
maximum twist in the shaft
maximum shear stress in the shaft
minimum twist in the shaft
A twist of one radian per unit length of the shaft.

ANSWER: D

Strut is defined as a
member of a structure which carries a tensile load
member of a structure which carries an axial compressive load
vertical member of a structure which carries a tensile load
None of the above.

ANSWER: B

Column is defined as a
member of a structure which carries a tensile load
member of a structure which carries an axial compressive load
vertical member of a structure which carries a tensile load
Vertical member of a structure which carries an axial compressive load.

ANSWER: D

The maximum axial compressive load which a column can take without failure by lateral deflection is called
critical load
buckling load
crippling load
Any one of the above.

ANSWER: D

Slenderness ratio is defined as the ratio of
equivalent length of the column to the minimum radius of gyration
length of the column to the minimum radius of gyration
length of the column to the area of cross-section of the column
Minimum radius of gyration to the area of cross-section of the column.

ANSWER: B

Buckling factor is defined as the ratio of

equivalent length of a column to the minimum radius of gyration
length of the column to the minimum radius of gyration
length of the column to the area of cross-section of the column
None of the above.

ANSWER: A

A loaded column is having the tendency to deflect. On account of this tendency, the critical load
decreases with the decrease in length
decreases with the increase in length
first decreases then increases with the decrease in length
First increases then decreases with the decrease in length.

ANSWER: B

A loaded column fails due to
stress due to direct load
stress due to bending
both (a) and (b)
None of the above.

ANSWER: C

The type of mechanical component that has a circular cross section and is subjected to a torque, on both ends is called:

- A. bar
- B. beam
- C. shaft
- D. column

ANSWER: C

when a mechanical component is subjected to a series of external axial loads that pass through the centroid of the cross section, and result in change in length in the material then the type of the element would be:

- A. bar
- B. beam
- C. shaft
- D. column

ANSWER: A

When the external load is applied in the transverse direction to the longitudinal axis of the component and causes the element to flex/bend due to the applied load, then the type of the element can be identified as:

- A. bar
- B. beam
- C. shaft
- D. column

ANSWER: B

The resistance per unit area, offered by a body against deformation is known as .

- A. stress
- B. strain
- C. elongation
- D. deflection

ANSWER: A

Which one of the following units of measurements cannot express stress?

- A. N/m^2
- B. 10^{-4} N/cm^2
- C. 10^{-6} N/mm^2
- D. 10^6 N/mm^2

ANSWER: D

The ratio of change of dimension of the body to the original dimension is known as

- A. stress
- B. strain
- C. elongation
- D. deflection

ANSWER: B

The stress induced in a body, which is subjected to two equal and opposite pulls, is known as

- A. compressive stress
- B. tensile strain
- C. tensile stress
- D. bearing stresses

ANSWER: C

The stress induced in a body, which is subjected to two equal and opposite pushes, is known as

- A. compressive stress
- B. tensile strain
- C. tensile stress
- D. bearing stress

ANSWER: A

is the property by virtue of which certain materials return back to their original position after the removal of the external force.

- A. plasticity
- B. elasticity
- C. ductility
- D. machinability

ANSWER: B

Which one of the following is wrong about Hooke's law:

- A. the stress is proportional to the strain.
- B. the shear stress is proportional to the shear strain.
- C. it is applicable only within elastic limit
- D. it does work within the plastic region

ANSWER: D

The ratio of tensile or compressive stress to the corresponding strain is known as

- A. Young's modulus
- B. modulus of rigidity
- C. shear modulus
- D. modulus of toughness

ANSWER: A

The stresses induced in a body due to change in temperature are known as

- A. thermal stress
- B. tensile strain
- C. shear stress
- D. bearing stresses

ANSWER: A

A rod 200 cm long and of diameter 3.0 cm is subjected to an axial pull of 30 kN. If the Young's modulus of the material of the rod is 2×10^5 N/mm², determine: the stress?

- A. 42.44 N/mm²
- B. 56.04 N/mm²
- C. 49.21 N/mm²
- D. 64.24 N/mm²

ANSWER: A

Find the Young's modulus of a rod of diameter 30 mm and of length 300 mm which is subjected

to a tensile load of 60 kN and the extension of the rod is equal to 0.4 mm.

- A. $45 \times 10^9 \text{ N/mm}^2$
- B. $63.6 \times 10^9 \text{ N/mm}^2$
- C. $56.6 \times 10^6 \text{ N/mm}^2$
- D. 84.24 N/mm^2

ANSWER: B

The extension in a rectangular steel bar of length 800 mm and of thickness 20 mm, is found to be 0.21 mm. The bar tapers uniformly in width from 80 mm to 40 mm. If E for the bar is $2 \times 10^5 \text{ N/mm}^2$, determine the axial tensile load on the bar.

- A. 45 kN
- B. 60.6 kN
- C. 84.7 kN
- D. 96 kN

ANSWER: B

A rod is 3 m long at a temperature of 15°C . Find the expansion of the rod, when the temperature is raised to 95°C . Take $E = 1 \times 10^5 \text{ N/mm}^2$ and $\alpha = 0.000012$ per degree centigrade.

- A. 1.35 cm
- B. 1.35 m
- C. 0.288 m
- D. 0.288 cm

ANSWER: D

When a body is subjected to an axial tensile load

- A. there is an increase in the width of the body
- B. there is an increase in the length of the body
- C. there is an increase in other dimensions of the body at right angles to the axis of the body
- D. no change in dimension at all

ANSWER: B

Which one is not correct?

- A. Poisson's ratio is the ratio of lateral strain to the longitudinal strain
- B. If longitudinal strain is tensile, the lateral strains will be compressive.
- C. lateral strain is opposite in sign to longitudinal strain
- D. lateral strain is always the same in sign to longitudinal strain

ANSWER: D

When an element is subjected to a set of shear stresses

- A. The planes of maximum normal stresses are perpendicular to each other.
- B. The planes of maximum normal stresses are inclined at an angle of 45° to the planes of pure shear.
- C. One of the maximum normal stress is tensile while the other maximum normal stress is compressive.
- D. All

ANSWER: D

The planes, which have no shear stress, are known as

- A. inclined plane
- B. principal plane
- C. oblique plane
- D. slanted plane

ANSWER: B

The angle made by the resultant stress with the normal of the oblique plane, is known as

- A. angle of twist
- B. obliquity
- C. straight line angle
- D. none

ANSWER: B

Maximum shear stress by Mohr's circle method, is equal to

- A. The radius of the Mohr's circle.
- B. The diameter of the Mohr's circle.
- C. maximum normal stress
- D. resultant stress

ANSWER: A

The maximum energy stored in the body without permanent deformation (i.e., upto elastic limit) is known as

- A. Resilience
- B. Proof Resilience.
- C. Modulus of resilience
- D. strain energy

ANSWER: B

A beam which is fixed at one end and free at the other end, is known as

- A. Simply supported beam
- B. Fixed beams
- C. Cantilever beam
- D. Overhanging beam

ANSWER: C

A beam which is provided more than two supports is known as

- A. Simply supported beam
- B. Fixed beams
- C. Continuous Beam
- D. Overhanging beam

ANSWER: C

Which one of the following is wrong about the assumptions made in the theory of simple bending?

- A. The material of the beam is homogeneous and isotropic
- B. The transverse sections which were plane before bending, remain plane after bending also.
- C. The value of young's modulus of elasticity is the same in tension and compression.
- D. The radius of curvature is small when compared with the dimensions of the cross-section.

ANSWER: D

if a section of a beam is subjected to a transversal load applied to the beam from above, and Cause pure bending in the beam and the longitudinal beam concaves upward, then the stress variation will be:

- A. The stress at the neutral axis is zero.
- B. the stresses will be compressive at any point above the neutral axis
- C. the stresses will be tensile below the neutral axis.
- D. all

ANSWER: D

Within elastic limit in a loaded material, stress is

- A. inversely proportional to strain
- B. Adirectly proportional to strain
- C. equal to strain
- D. none of the above.

ANSWER: B

The ratio of linear stress to linear strain is known as

- A. Poisson's ratio
- B. bulk modulus

- C. modulus of rigidity
- D. modulus of elasticity.

ANSWER: D

The ratio of normal stress of each face of a solid cube to volumetric strain is called

- A. Poisson's ratio
- B. bulk modulus
- C. modulus of rigidity
- D. modulus of elasticity.

ANSWER: B

Hooke's law holds good upto

- A. proportional limit
- B. yield point
- C. elastic limit
- D. plastic limit.

ANSWER: A

The property of a material by virtue of which a body returns to its original shape after removal of the load is known as

- A. ductility
- B. plasticity
- C. elasticity
- D. resilience.

ANSWER: C

The work done in producing strain on a material per unit volume is called

- A. resilience
- B. ductility
- C. elasticity
- D. plasticity

ANSWER: A

The property by virtue of which a metal can be beaten into plates is called

- A. ductility
- B. malleability
- C. resilience
- D. plasticity.

ANSWER: B

The stress due to suddenly applied load as compared to the stress due to the same load gradually applied to the same rod is

- A. half
- B. same

- C. double
- D. three times.

ANSWER: C

If a member, whose tensile strength is more than two times the shear strength, is subjected to an axial load upto failure, the failure of the member will occur by

- A. maximum normal stress
- B. maximum shear stress
- C. normal stress or shear stress
- D. none of the above.

ANSWER: B

If a beam is fixed at both its ends, it is called a

- A. fixed beam
- B. built-in beam
- C. encastered beam
- D. any one of the above

ANSWER: D

If a beam is supported on more than two supports, it is called a

- A. built-in beam
- B. continuous beam
- C. simply supported beam
- D. encastered beam.

ANSWER: B

A cantilever of length (l) carries a point load (W) at the free end. The bending moment diagram will be a

- A. parabola with maximum ordinate at the center of the beam
- B. parabola with maximum ordinate at the cantilever end
- C. triangle with maximum ordinate at the free end
- D. triangle with maximum ordinate at the cantilever end.

ANSWER: D

A cantilever of length (l) carries a uniformly distributed load over the whole length. The bending moment diagram will be

- A. parabola with maximum ordinate at the center
- B. parabola with maximum ordinate at the cantilever end
- C. triangle with maximum ordinate at the free end
- D. triangle with maximum ordinate at the cantilever end.

ANSWER: B

A simply supported beam of span (l) carries a uniformly distributed load over the whole span. The shear force diagram will be

- A. a rectangle
- B. a triangle
- C. two equal and opposite rectangles
- D. two equal and opposite triangles.

ANSWER: D

The bending moment on a section is maximum where shearing force is

- A. minimum
- B. maximum
- C. zero
- D. changing sign

ANSWER: D

The point of contraflexure occurs only in

- A. continuous beams
- B. cantilever beams
- C. overhanging beams
- D. simply supported beams

ANSWER: C

The bending moment at a section, where shear force is zero, will be

- A. zero
- B. maximum
- C. minimum
- D. either minimum or maximum

ANSWER: D

The shear force and bending moment are zero at the free end of a cantilever, if it carries a

- A. point load at the free end
- B. uniformly distributed load over the whole length
- C. point load in the middle of its length
- D. none of the above.

ANSWER: B

In a tensile test of a specimen, the ratio of maximum load to the original cross-sectional area of the test piece is called

- A. yield stress
- B. ultimate stress
- C. safe stress
- D. breaking stress.

ANSWER: B

A solid circular shaft of diameter D carries an axial load W . If the same load is applied axially on a hollow circular shaft of inner diameter as $D/2$, the ratio of stresses in a solid shaft to that of

hollow shaft would be

- A. $\frac{1}{2}$
- B. $\frac{1}{4}$
- C. $\frac{4}{3}$
- D. $\frac{3}{4}$

ANSWER: D

The diameter of a mild steel round bar, on which tensile test is performed, at fracture will

- A. increase
- B. decrease
- C. be same
- D. none of the above.

ANSWER: B

A laminated spring 1 m long carries a central point load of 2000 N. The spring is made up of plates each 5 cm wide and 1 cm thick. The bending stress in the plates is limited to 10 N/mm². The number of plates required, will be

- A. 3
- B. 5
- C. 6
- D. 8.

ANSWER: C

A fixed beam is a beam whose end supports are such that the end slopes

- A. are maximum
- B. are minimum
- C. are zero
- D. none of the above.

ANSWER: C

The ratio of moment of inertia about the neutral axis to the distance of the most distance point of the section from the neutral axis is called

- A. moment of inertia
- B. section modulus
- C. polar moment of inertia
- D. Modulus of rigidity.

ANSWER: B

Fletched beam means a

- A. continuous beam
- B. fixed beam
- C. beam of composite section consisting of a wooden beam strengthened by mild steel plates
- D. None of the above.

ANSWER: C

The shear stress required to cause plastic deformation of solid metal is called

- A. proof stress
- B. flow stress
- C. rupture stress
- D. Ultimate stress.

ANSWER: B

The stress which will cause a specified permanent deformation in a material (usually 0.01% or less) is called

- A. proof stress
- B. flow stress
- C. rupture stress
- D. Ultimate stress.

ANSWER: A

The stress obtained by dividing the load at the moment of incipient fracture, by the area supporting that load is called

- A. proof stress
- B. flow stress
- C. rupture stress
- D. Ultimate stress.

ANSWER: C

A short column of rectangular section carries a point load (W) acting with an eccentricity (e). The shape of Kernel area would be

- A. square
- B. rectangle
- C. circle
- D. Rhombus.

ANSWER: D

If in question 79, the column is of circular section then the shape of Kernel area would be

- A. square
- B. rectangle
- C. circle
- D. Rhombus.

ANSWER: C

Every cross-section of a shaft, which is subjected to a twisting moment, is under

- A. compressive stress
- B. shear stress
- C. tensile stress
- D. Bending stress.

ANSWER: B

The shear stress at any point of a shaft, subjected to twisting moment, is

- A. proportional to its distance from the central axis of the shaft
- B. inversely proportional to its distance from the central axis of the shaft
- C. proportional to the square of its distance from the central axis of the shaft
- D. None of the above.

ANSWER: A

Torsional rigidity of a shaft is equal to

- A. product of modulus of rigidity and polar moment of inertia
- B. sum of modulus of rigidity and polar moment of inertia
- C. difference of modulus of rigidity and polar moment of inertia
- D. Ratio of modulus of rigidity and polar moment of inertia.

ANSWER: A

Polar modulus of a shaft section is equal to

- A. product of polar moment of inertia and maximum radius of the shaft
- B. ratio of polar moment of inertia to maximum radius of the shaft
- C. sum of polar moment of inertia and maximum radius of the shaft
- D. Difference of polar moment of inertia and maximum radius of the shaft.

ANSWER: B

The torsional rigidity of a shaft is defined as the torque required to produce

- A. maximum twist in the shaft
- B. maximum shear stress in the shaft
- C. minimum twist in the shaft
- D. A twist of one radian per unit length of the shaft.

ANSWER: D

Strut is defined as a

- A. member of a structure which carries a tensile load
- B. member of a structure which carries an axial compressive load
- C. vertical member of a structure which carries a tensile load
- D. None of the above.

ANSWER: B

Column is defined as a

- A. member of a structure which carries a tensile load
- B. member of a structure which carries an axial compressive load
- C. vertical member of a structure which carries a tensile load
- D. Vertical member of a structure which carries an axial compressive load.

ANSWER: D

The maximum axial compressive load which a column can take without failure by lateral deflection is called

- A. critical load
- B. buckling load
- C. crippling load
- D. Any one of the above.

ANSWER: D

Slenderness ratio is defined as the ratio of

- A. equivalent length of the column to the minimum radius of gyration
- B. length of the column to the minimum radius of gyration
- C. length of the column to the area of cross-section of the column
- D. Minimum radius of gyration to the area of cross-section of the column.

ANSWER: B

Buckling factor is defined as the ratio of

- A. equivalent length of a column to the minimum radius of gyration
- B. length of the column to the minimum radius of gyration
- C. length of the column to the area of cross-section of the column
- D. None of the above.

ANSWER: A

A loaded column is having the tendency to deflect. On account of this tendency, the critical load

- A. decreases with the decrease in length
- B. decreases with the increase in length
- C. first decreases then increases with the decrease in length
- D. First increases then decreases with the decrease in length.

ANSWER: B

A loaded column fails due to

- A. stress due to direct load
- B. stress due to bending
- C. both (a) and (b)
- D. None of the above.

ANSWER: C

The type of mechanical component that has a circular cross section and is subjected to a torque, on both ends is called:

- A. bar
- B. beam
- C. shaft
- D. column

ANSWER: C

when a mechanical component is subjected to a series of external axial loads that pass through the centroid of the cross section, and result in change in length in the material then the type of the element would be:

- A. bar
- B. beam
- C. shaft
- D. column

ANSWER: A

When the external load is applied in the transverse direction to the longitudinal axis of the component and causes the element to flex/bend due to the applied load, then the type of the element can be identified as:

- A. bar
- B. beam
- C. shaft
- D. column

ANSWER: B

The resistance per unit area, offered by a body against deformation is known as .

- A. stress
- B. strain
- C. elongation
- D. deflection

ANSWER: A

Which one of the following units of measurements cannot express stress?

- A. N/m^2
- B. 10^{-4} N/cm^2
- C. 10^{-6} N/mm^2
- D. 10^6 N/mm^2

ANSWER: D

The ratio of change of dimension of the body to the original dimension is known as

- A. stress
- B. strain
- C. elongation
- D. deflection

ANSWER: B

The stress induced in a body, which is subjected to two equal and opposite pulls, is known as

- A. compressive stress
- B. tensile strain
- C. tensile stress
- D. bearing stresses

ANSWER: C

The stress induced in a body, which is subjected to two equal and opposite pushes, is known as

- A. compressive stress
- B. tensile strain
- C. tensile stress
- D. bearing stress

ANSWER: A

is the property by virtue of which certain materials return back to their original position after the removal of the external force.

- A. plasticity
- B. elasticity
- C. ductility
- D. machinability

ANSWER: B

Which one of the following is wrong about Hooke's law:

- A. the stress is proportional to the strain.
- B. the shear stress is proportional to the shear strain.
- C. it is applicable only within elastic limit
- D. it does work within the plastic region

ANSWER: D

The ratio of tensile or compressive stress to the corresponding strain is known as

- A. Young's modulus
- B. modulus of rigidity
- C. shear modulus
- D. modulus of toughness

ANSWER: A

The stresses induced in a body due to change in temperature are known as

- A. thermal stress
- B. tensile strain
- C. shear stress
- D. bearing stresses

ANSWER: A

A rod 200 cm long and of diameter 3.0 cm is subjected to an axial pull of 30 kN. If the Young's modulus of the material of the rod is 2×10^5 N/mm², determine: the stress?

- A. 42.44 N/mm²
- B. 56.04 N/mm²
- C. 49.21 N/mm²
- D. 64.24 N/mm²

ANSWER: A

Find the Young's modulus of a rod of diameter 30 mm and of length 300 mm which is subjected to a tensile load of 60 kN and the extension of the rod is equal to 0.4 mm.

- A. 45×10^9 N/mm²
- B. 63.6×10^9 N/mm²
- C. 56.6×10^6 N/mm²
- D. 84.24 N/mm²

ANSWER: B

The extension in a rectangular steel bar of length 800 mm and of thickness 20 mm, is found to be 0.21 mm. The bar tapers uniformly in width from 80 mm to 40 mm. If E for the bar is 2×10^5 N/mm², determine the axial tensile load on the bar.

- A. 45 kN
- B. 60.6 kN
- C. 84.7 kN
- D. 96 kN

ANSWER: B

A rod is 3 m long at a temperature of 15°C. Find the expansion of the rod, when the temperature is raised to 95°C. Take $E = 1 \times 10^5$ N/mm² and $\alpha = 0.000012$ per degree centigrade.

- A. 1.35 cm
- B. 1.35 m
- C. 0.288 m
- D. 0.288 cm

ANSWER: D

When a body is subjected to an axial tensile load

- A. there is an increase in the width of the body
- B. there is an increase in the length of the body
- C. there is an increase in other dimensions of the body at right angles to the axis of the body
- D. no change in dimension at all

ANSWER: B

Which one is not correct?

- A. Poisson's ratio is the ratio of lateral strain to the longitudinal strain
- B. If longitudinal strain is tensile, the lateral strains will be compressive.
- C. lateral strain is opposite in sign to longitudinal strain
- D. lateral strain is always the same in sign to longitudinal strain

ANSWER: D

When an element is subjected to a set of shear stresses

- A. The planes of maximum normal stresses are perpendicular to each other.
- B. The planes of maximum normal stresses are inclined at an angle of 45° to the planes of pure

shear.

C. One of the maximum normal stress is tensile while the other maximum normal stress is compressive.

D. All

ANSWER: D

The planes, which have no shear stress, are known as

A. inclined plane

B. principal plane

C. oblique plane

D. slanted plane

ANSWER: B

The angle made by the resultant stress with the normal of the oblique plane, is known as

A. angle of twist

B. obliquity

C. straight line angle

D. none

ANSWER: B

Maximum shear stress by Mohr's circle method, is equal to

A. The radius of the Mohr's circle.

B. The diameter of the Mohr's circle.

C. maximum normal stress

D. resultant stress

ANSWER: A

The maximum energy stored in the body without permanent deformation (i.e., upto elastic limit) is known as

A. Resilience

B. Proof Resilience.

C. Modulus of resilience

D. strain energy

ANSWER: B

A beam which is fixed at one end and free at the other end, is known as

A. Simply supported beam

B. Fixed beams

C. Cantilever beam

D. Overhanging beam

ANSWER: C

A beam which is provided more than two supports is known as

A. Simply supported beam

- B. Fixed beams
- C. Continuous Beam
- D. Overhanging beam

ANSWER: C

Which one of the following is wrong about the assumptions made in the theory of simple bending?

- A. The material of the beam is homogeneous and isotropic
- B. The transverse sections which were plane before bending, remain plane after bending also.
- C. The value of young's modulus of elasticity is the same in tension and compression.
- D. The radius of curvature is small when compared with the dimensions of the cross-section.

ANSWER: D

if a section of a beam is subjected to a transversal load applied to the beam from above, and Cause pure bending in the beam and the longitudinal beam concaves upward, then the stress variation will be:

- A. The stress at the neutral axis is zero.
- B. the stresses will be compressive at any point above the neutral axis
- C. the stresses will be tensile below the neutral axis.
- D. all

ANSWER: D

_____ of a gas is the energy stored in it by virtue of its molecular motion.

- A. Specific heat
- B. Internal energy
- C. Enthalpy
- D. Mach number

ANSWER: B

_____ were employed to obtain pressure rise in flowing gases for a given value of initial kinetic energy

- A. Nozzle
- B. Impeller
- C. Guide vane
- D. Diffuser

ANSWER: D

In the case of pumps with equal diameters of suction and pressure flanges, the velocity energy is _____

- A. High
- B. Low
- C. zero
- D. Constant

ANSWER: C

A pair of fixed blade and rotor blade in axial flow compressor is called as _____.

- A. Step
- B. Pair
- C. State
- D. Stage

ANSWER: D

_____ can be defined as the ratio of the pressure rise in rotor blades to the pressure rise in stages in an axial flow compressor

- A. Degree of reaction
- B. Degree of pressure
- C. Pressure ratio
- D. Reaction ratio

ANSWER: A

The degree of reaction is usually kept _____ for all types of axial flow compressors.

- A. 0.2
- B. 0.5
- C. 0.4
- D. 0.3

ANSWER: B

Discharge capacity of the reciprocating pump is _____ that of the centrifugal pump.

- A. lower than
- B. higher than
- C. same as
- D. unpredictable

ANSWER: A

The jet ratio for pelton wheel turbine is _____

- A. 14
- B. 18
- C. 12
- D. 10

ANSWER: C

Francis turbine is _____

- A. Radial flow turbine
- B. Axial flow turbine
- C. Mixed flow turbine
- D. Inward flow radial type turbine

ANSWER: D

The diverging portion of the nozzle acts as a diffuser for_____

- A. the subsonic flow
- B. the supersonic flow
- C. both the subsonic as well as supersonic flow
- D. none of the above

ANSWER: A